

Digital Signal Processing for Short-Reach Optical Communications: A Review of Current Technologies and Future Trends

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(Invited Paper)

Abstract—Driven primarily by cloud service and data-center applications, short-reach optical communication has become a key market segment and growing research area in recent years. Short-reach systems are characterized by direct detection-based receiver configurations and other low-cost and small form factor components that induce transmission impairments unforeseen in their coherent counterparts. Innovative signaling and digital signal processing (DSP) play a pivotal role in enabling these components to realize their ultimate potentials and meet data rate requirements in cost-effective manners. This paper presents an overview of recent DSP developments for short-reach communications systems and discusses future trends.

Index Terms—Digital signal processing, direct detection, optical communications.

I. INTRODUCTION

FIBER-OPTIC communications traditionally focused on high speed connections over vast distances across the globe. In recent years, new applications such as Internet of Things (IoT), Augmented/Virtual Reality, Cloud Computing, storage and other software-as-a-service (SaaS) and Infrastructure-as-a-Service (IaaS) have driven demands for short-reach links in data centers (DCs). Short-reach communications collectively refer to optical communication links in the hundreds

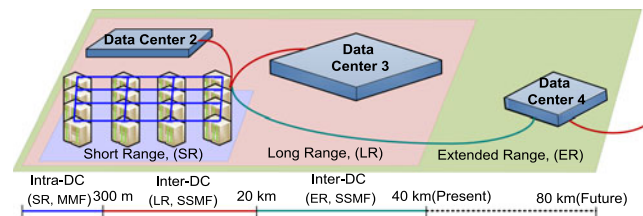


Fig. 1. The length scale of short-reach optical communications for DCs can range from <300 m using multi-mode fiber to over 20 km using single-mode fibers with advanced modulation formats.

of meters connecting one server to another within a DC to inter-DC links in the tens of kilometers as shown in Fig. 1. Note that such extended reach (ER) inter-DC connections have length scales approaching typical Metro and Access links. On the other extreme, on-board chip-to-chip optical interconnects are links with even shorter distance. Their design principles and considerations such as electronic packaging, power, thermal management and other mechanical aspects are more Application Specific Integrated Circuit (ASIC) development related and are not the focus of this review paper. Interested readers are referred to [1] for a detailed discussion on the topic.

A recent Cisco report predicts that 86% of the global Internet traffic will be DC related in which 77% will be within a DC by 2020. Such explosive growth for DC traffic is driven by content providers such as Facebook, Google, Amazon and the like. Over the past few years, they have heavily invested in their own infrastructure and overtaken traditional telecommunication carriers as market drivers for optical transceivers especially for short-reach links.

In terms of developments in optical transceiver technologies, perhaps the most significant advancement in the past decade is digital coherent systems in which all degrees of freedom of the optical carrier (amplitude, phase, polarization) are exploited to encode information. Receiver digital signal processing (DSP) recovers the full field information in both polarizations. Transmission impairments like chromatic dispersion (CD) and polarization-mode dispersion (PMD) become linear and can be compensated easily and adaptively. Today, high order modulation formats such as polarization-division multiplexing-quadrature phase shift keying (PDM-QPSK) and polarization-division multiplexing-16 quadrature amplitude modulation

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(PDM-16QAM) enable transmission bit rates per channel up to 100 Gbit/s and 200 Gbit/s, respectively. Single-channel 1 Tbit/s transmission was recently demonstrated [2] and further developments into commercial transceivers are expected in the near future.

In recent years, the research community has put tremendous efforts to develop DSP strategies for short-reach systems to mirror the achievements in long-haul systems. However, the sheer market size for short reach transceivers renders cost the prime concern and prohibits direct application of the digital coherent system philosophy into short-reach links. Also, transceiver form factors are particularly important as port density, or total number of physical connections and bit rate per area, is a key parameter determining the total size and energy usage of DC. These constraints impose unprecedented challenges in physical layer devices and necessitate innovative signaling, detection and DSP strategies to realize next generation DC for various applications aforementioned.

In this paper, we extend our discussion from our OFC contribution [3] and provide a more comprehensive review on this important topic of DSP for short-reach communication systems. The rest of the paper is organized as follows: Section II reviews the background of signaling techniques and DSP for coherent systems and highlights unique challenges for short-reach systems. This will be followed by modulation formats for short-reach systems in Section III. Section IV describes key DSP techniques at the receiver and Section V extends the discussions to PDM transmission with direct-detection receivers. Device advances for short-reach transceivers and their impacts on DSP development are discussed in Section VI. Conclusions and future outlook will be given in Section VII.

II. CHANNEL MODEL AND DSP FOR LONG-HAUL VS SHORT-REACH SYSTEMS

The current wave of DSP research for short-reach links is based on advances in digital coherent systems for long-haul systems over the past decade. We will first provide an overview of coherent detection and DSP for long-haul systems followed by a discussion of constraints specific to short-reach systems which lead to new DSP research directions.

A. Canonical Digital Coherent Systems for Long-Haul Optical Communications

Consider a full polarization-division-multiplexed coherent system. Let $E_{t,x(y)}(t)$ denote the transmitted signal in the $x - (y-)$ polarization given by

$$E_{t,x(y)}(t) = \sum_k b_{k,x(y)} p(t - kT_s) \quad (1)$$

where $b_{k,x(y)}$ are the k th information symbols in the $x - (y-)$ polarization, $p(t)$ is the pulse shape and T_s is the symbol period, respectively. In a long-haul fiber-optic channel, major transmission impairments include [4], [5]: a) CD characterized by the transfer function $H_{CD}(\omega) = e^{-j\omega^2 \beta_2 L/2}$; b) first-order PMD

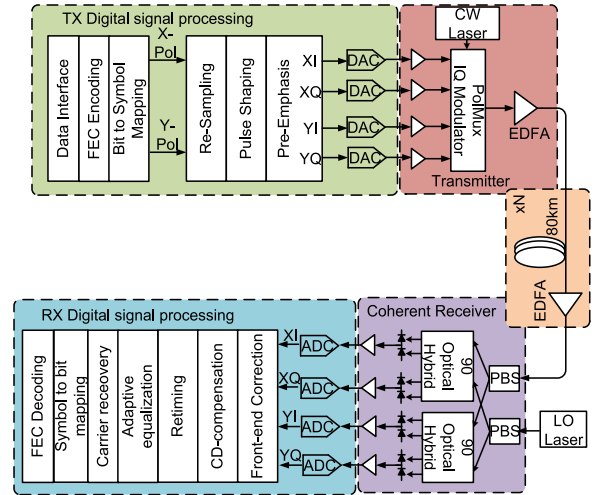


Fig. 2. Standard digital coherent transceiver system with typical DSP configurations.

characterized by the 2×2 PMD matrix

$$H_{PMD}(\omega) = \begin{bmatrix} e^{j\frac{\epsilon_1}{2}} \cos \frac{\theta_1}{2} & -e^{-j\frac{\epsilon_1}{2}} \sin \frac{\theta_1}{2} \\ e^{j\frac{\epsilon_1}{2}} \sin \frac{\theta_1}{2} & e^{-j\frac{\epsilon_1}{2}} \cos \frac{\theta_1}{2} \end{bmatrix} \begin{bmatrix} e^{j\frac{\omega\tau}{2}} & 0 \\ 0 & e^{-j\frac{\omega\tau}{2}} \end{bmatrix} \times \begin{bmatrix} e^{j\frac{\epsilon_2}{2}} \cos \frac{\theta_2}{2} & -e^{-j\frac{\epsilon_2}{2}} \sin \frac{\theta_2}{2} \\ e^{j\frac{\epsilon_2}{2}} \sin \frac{\theta_2}{2} & e^{-j\frac{\epsilon_2}{2}} \cos \frac{\theta_2}{2} \end{bmatrix} \quad (2)$$

where $\theta_1, \theta_2, \epsilon_1$ and ϵ_2 are the polar and azimuthal angles and τ is the differential group delay (DGD) and c) amplified spontaneous emission (ASE) noise $v_{x,ASE}(t), v_{y,ASE}(t)$ from inline optical amplifiers, which are typically modeled as complex circularly symmetric zero-mean additive white Gaussian random process. Note that laser imperfections such as laser phase noise and frequency offset between transmitter laser and local oscillators (LOs) are also important impairments for coherent systems but will be omitted here for brevity. Also, fiber Kerr nonlinearity is another fundamental impairment for long-haul systems, but its effect is mostly negligible for short distance transmissions and hence is not included in the discussion here. In this case, the received signal is given by

$$\begin{bmatrix} E_{r,x}(t) \\ E_{r,y}(t) \end{bmatrix} = \mathcal{F}^{-1} \left\{ H_{CD}(\omega) H_{PMD}(\omega) \begin{bmatrix} \tilde{E}_{t,x}(\omega) \\ \tilde{E}_{t,y}(\omega) \end{bmatrix} \right\} + \begin{bmatrix} v_{x,ASE}(t) \\ v_{y,ASE}(t) \end{bmatrix} \quad (3)$$

where $\tilde{E}_{t,x(y)}(\omega)$ is the Fourier transform of $E_{t,x(y)}(t)$ and $\mathcal{F}^{-1}$ denotes inverse Fourier transform. The fiber CD is a static impairment while PMD is time varying and caused by random mechanical perturbations and temperature fluctuations in the fiber at random locations. Interested readers are referred to [7] for an in-depth treatment of PMD in optical fibers.

A standard digital coherent transceiver system with typical DSP functions is shown in Fig. 2. It consists of three major blocks: Transmitter side DSP and electrical-to-optical (E/O) conversion, optical fiber link, coherent receiver and receiver side DSP. At transmitter side, the data is first processed by the

following DSP functions: forward error correction (FEC) encoding, bit to symbol mapping, re-sampling and pulse shaping. The processed data is sent to digital-to-analog converters (DAC) for the generation of baseband electrical signal. A polarization multiplexed IQ modulator modulates the information onto the optical carrier and is sent into the fiber where inline optical amplifiers are used to compensate the fiber loss. A standard polarization and phase diversity coherent receiver is also shown which consists of a LO laser, two 90 degree optical hybrids and four balanced photo-detectors. After coherent detection, the received signals are sampled by ADC and processed by the following DSP functions: 1) front-end correction for receiver optical hybrids imperfections; 2) CD compensation; 3) re-timing; 4) adaptive equalization for polarization de-multiplexing, residual CD, PMD and timing errors, 5) carrier frequency offset compensation and phase estimation and 6) FEC decoding. CD can be compensated using static equalizer either in frequency domain or time domain. Polarization de-multiplexing is achieved with state-of-polarization (SOP) tracking, PMD and residual inter-symbol interference (ISI) compensation (due to residual CD or transceiver components bandwidth limitation) through multiple-input-multiple-output (MIMO) DSP techniques. Interested readers are referred to relevant review papers [4], [5] for a detailed treatment of the subject of DSP for long-haul coherent systems.

B. Unique Challenges for Short-Reach Systems

The unique challenges for short-reach systems that motivate new DSP techniques are multi-fold.

Cost: For the low-cost short-reach transceivers market, high performance devices such as low-linewidth external cavity lasers (ECL), Lithium Niobate (LiNbO₃) external modulators, thermoelectric cooling (TEC), LOs and full coherent receivers are prohibited. Instead, components such as vertical-cavity surface-emitting laser (VCSEL), directly modulated lasers (DML), electro-absorption modulators (EAM), low bandwidth PIN make up the building blocks of most short-reach transceivers. Also, direct detection is much preferred over coherent detection. Simplified DSP is much desired and channel spacing (for WDM implementation) is much larger than standard 50 GHz international telecommunication union (ITU) grids because of the lack of TEC and wavelength locking mechanisms.

Form factor: While small form factors are desired for all types of optical transceivers in general, the form factor is imperative for DC applications as it significantly affects the port density, rack size which directly translates into the overall size of DCs, the amount of cooling/air conditioning required and hence ultimately total energy usage.

Latency: For DC systems, latency tolerance is more stringent than long-haul systems because of super-computing and various other delay-sensitive Cloud applications. Also, propagation delay is so much smaller than long-haul links and there are fewer link components in short-reach systems to induce further signal delays. In this case, typical FEC latency of 15 to 150 μ s [6] become a major contributor of overall system latency and therefore complicated soft-decision FEC with iterative decoding

techniques is out of consideration. Standard hard decision-forward error correction (HD-FEC) or innovative low-latency codes are more relevant for short-reach systems. This in turn imposes more stringent pre-FEC bit error rate (BER) and receiver sensitivity (RS) requirements.

Low-cost components produce non-idealities resulting in new transmission impairments specific to short-reach systems. For example, directly modulating a DML will produce signal chirp given by $\Delta f = \frac{\alpha}{4\pi} (\frac{d}{dt} \ln P(t) + \kappa P(t))$ where $P(t)$ is the instantaneous optical power driven by the input signal (current) into the laser, α denotes laser linewidth enhancement factor and κ denotes adiabatic chirp coefficient. Other components like EAM, PD, and trans-impedance amplifier (TIA) also produce additional nonlinear distortions. Another common impairment in short-reach systems is considerable low-pass filtering effects due to insufficient bandwidth of various components. Furthermore, finite extinction ratios of EAM and Relative Intensity Noise (RIN) degrade signal-to-noise ratios (SNR) and reduce link power budget margins. Finally, direct detection with non-negative signaling complicates the equalization of linear transmission impairments like CD and PMD as one cannot simply multiply the received signal spectrum by the inverse of the CD/PMD transfer function. Moreover, most research works in short-reach systems consider polarization rotation only and neglect PMD effects because of the transmission length scale concerned. In this case, $H_{PMD}(\omega)$ reduces to a simple rotation matrix of the signal SOP

$$H_{PMD}(\omega) = H_{SOP} = \begin{bmatrix} e^{j\frac{\theta}{2}} \cos \frac{\theta}{2} & -e^{-j\frac{\theta}{2}} \sin \frac{\theta}{2} \\ e^{j\frac{\theta}{2}} \sin \frac{\theta}{2} & e^{-j\frac{\theta}{2}} \cos \frac{\theta}{2} \end{bmatrix} \quad (4)$$

where θ and ε are the polar and azimuthal angle. If we assume a simple polarization-demultiplexed direct detection receiver without optical amplifiers, the received photo-current $i_{r,x(y)}(t)$ will be given by

$$\begin{aligned} \begin{bmatrix} i_{r,x}(t) \\ i_{r,y}(t) \end{bmatrix} &= \begin{bmatrix} |E_{r,x}(t)|^2 \\ |E_{r,y}(t)|^2 \end{bmatrix} \\ &= \left| \mathcal{F}^{-1} \left\{ H_{CD}(\omega) H_{SOP} H_{LP}(\omega) \begin{bmatrix} \tilde{E}_{t,x}(\omega) \\ \tilde{E}_{t,y}(\omega) \end{bmatrix} \right\} \right|^2 \\ &\quad + \begin{bmatrix} v_{x,Rx}(t) \\ v_{y,Rx}(t) \end{bmatrix} \end{aligned} \quad (5)$$

where $H_{LP}(\omega)$ denotes low-pass filtering and the transmitted electric field $E_{t,x(y)}(t) = g(i_{t,x(y)}(t))$ is generally a nonlinear function $g(\cdot)$ of the input driving signal $i_{t,x(y)}(t)$ that models various lasers and modulator imperfections. Note that the noise is no longer amplifier noise but rather shot or thermal noise from photo-receivers.

We can categorize different short-reach systems according to their respective technology features and challenges:

- 1) *Server-to-server or intra-data-center links:* These are links below 300 m connecting one server to another within a data center. They constitute the biggest volume among all market segments in data center interconnects. They are currently dominated by VCSEL transmitters and MMF.

TABLE I
DIFFERENT TYPES OF SHORT-REACH SYSTEMS AND THEIR RESPECTIVE CHALLENGES

Type	Length scale	Features and Limiting factors	Remarks
Server-to-server, intra-data center, intra/inter-rack	< 300 m	Inter-modal dispersion for MMF	
Inter-data center	< 20 km	RS, CD	O-band transmissions as alternative
ER inter-data center, metro and access	< 80 km	CD	Fully coherent transceivers eventually?

- 2) *Inter data-center links*: These are links below 20 km connecting one data center to another. The length scale is too long for multi-mode fiber (MMF) because of inter-modal dispersion and hence standard single-mode fiber (SSMF) is the default choice. On the other hand, this length scale is still short enough that optical amplifiers are not preferred. Consequently, RS is an important parameter for system optimization. Advanced photo-diodes such as high-bandwidth avalanche photo-diodes (APD) enables much lower RS than traditional PIN and are important research achievements in this area. In addition, as CD effects grow with the baud rate and distance and direct detection eliminates signal phase information and make the system nonlinear, CD effects will need to be addressed by innovative signaling and DSP techniques. Alternatively, O-band transmissions are proposed to address such problems as CD effects are minimal at that wavelength range for SSMF.
- 3) *Extended reach inter-data-center, access and metro links*: These are links between 20 km and 80 km-long and optical amplification is acceptable for some scenarios. However, direct detection receivers are still preferred over fully coherent receivers. To further increase the bit rate, polarization-multiplexed direct detection configurations such as Stokes-vector direct detection (SV-DD) receivers are being actively pursued by various groups worldwide. On the other hand, there are ongoing efforts to push the bit rate and transmission distance of O-band systems without optical amplification into this application domain. Furthermore, as full coherent transceivers are becoming cheaper, smaller and more power efficient, extended range(ER) inter-data-center links may adopt fully coherent transceivers in the foreseeable future.

Table I summarizes the key features and limiting factors of the few types of short-reach systems aforementioned. We will review modulation formats and DSP techniques to meet the respective challenges for each type of short-reach systems in the following sections.

III. MODULATION FORMATS FOR SHORT REACH SYSTEMS

Intensity modulation combined with direct detection (IM/DD) systems, which are technologically simple, cost effective and small form-factor, become the most practical solution

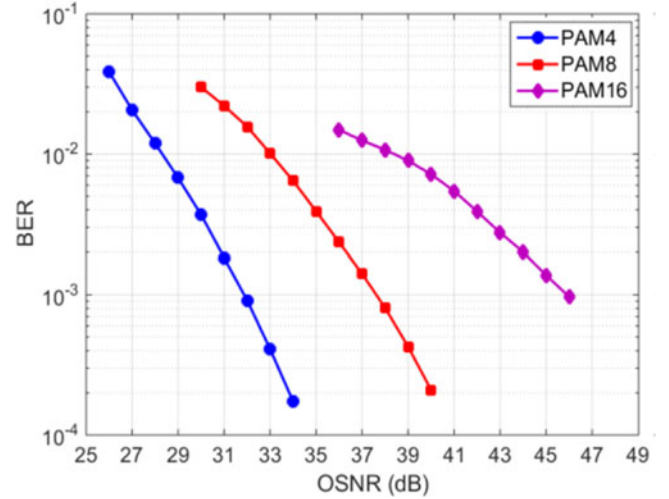


Fig. 3. Simulation results for BER vs. optical signal-to-noise ratio (OSNR) for 112 Gbit/s PAM-4, PAM-8 and PAM-16.

for short-reach systems [8]. Traditional optical interconnects implemented with simple non-return to zero-on-off-keying (NRZ-OOK) format and direct detection receivers without any special DSP (normally up to 25 Gbit/s) are no longer sufficient to support the increasing interface speed requirements. In order to scale transmission speed of short-reach systems, we will review common approaches to achieve high order modulation formats including pulse amplitude modulation (PAM), carrier-less amplitude and phase modulation (CAP) and discrete multi-tone (DMT) modulation.

PAM: PAM is the simplest high order modulation formats. Unlike NRZ-OOK format, information is encoded on multiple amplitude levels such as PAM-4, PAM-8 or PAM-16. As the number of levels doubles, the RS will be reduced significantly. Fig. 3 shows the BER vs. OSNR for 112 Gbit/s PAM-4, PAM-8, and PAM-16. It can be seen that higher OSNR is required for higher order PAM signal and are more sensitive to channel impairments such as CD and requires more complex DSP. Therefore, PAM-4 is a more practical option considering the trade-off between system performance and achievable bit rate. Recently, the IEEE 400GbE P802.3bs task force has adopted PAM-4 for data center interconnects [9].

Fig. 4 shows the DSP configuration for PAM transmission systems. Pulse shaping and pre-emphasis through a time-domain finite impulse response (FIR) filter is employed to compensate the DAC bandwidth limitations and transmitter nonlinearity such as the nonlinear characteristics of the modulator [11]. The signal is then loaded into DAC to generate the electrical driving signal, boosted by linear driver amplifier and modulate the laser. The laser can be a DML, electro-absorption modulators integrated with distributed feedback lasers (EML) or VCSEL. The received optical signal will be direct detected by PIN PD or APD. Compared to the receiver DSP of coherent systems, the difference of PAM-4 direct detection systems is the adaptive equalization using feed forward equalizer (FFE), decision feedback equalizer (DFE) to compensate for various transceiver and channel impairments [12], [13]. However, when the bandwidth

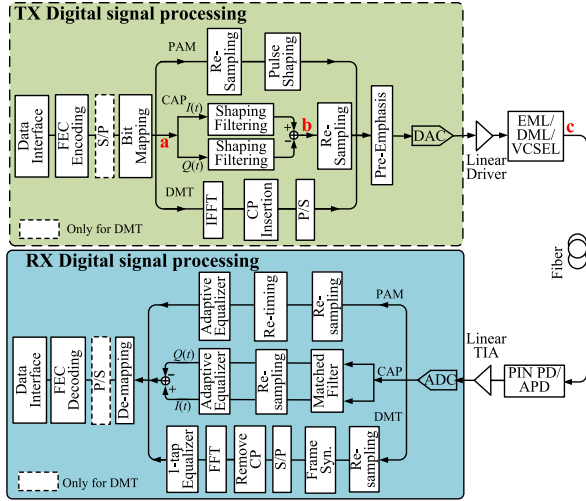


Fig. 4. DSP configuration for PAM/CAP/DMT transmission systems. The signal spectra at points a, b and c are shown in Fig. 5(b)–5(d) respectively.

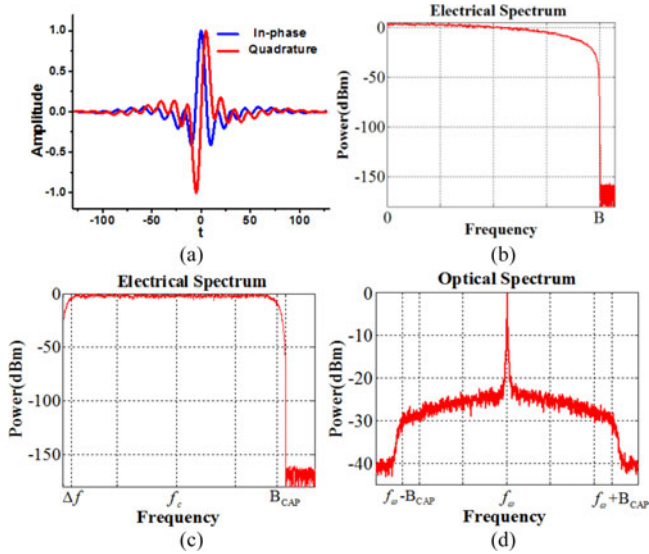


Fig. 5. (a) Response of shaping filter. (b) Electrical spectrum of baseband signal. (c) Electrical spectrum of electrical CAP signal. (d) Optical spectrum of CAP signal.

of the concatenated components in the link is considerably lower than signal bandwidth, significant ISI will be induced. Compensation of ISI by linear equalizers only will greatly enhance the noise in the high frequency region which are not desirable. Instead, techniques such as MLSE can be employed. In this connection, Zhong *et al.* proposed direct detection faster than Nyquist (DD-FTN) algorithm to overcome the drawbacks of conventional equalizers [14]. We will discuss these algorithms in more details in the next section.

CAP: An alternative intensity modulation format that can provide high order modulation is CAP modulation that enables the generation of QAM-type signals but with simpler implementation. In this case, baseband QAM signal with a frequency shift to a subcarrier is required. CAP uses FIR filter to achieve this up-conversion and modulate QAM signal to a real signal or demodulate QAM signal from a received real signal. Although the

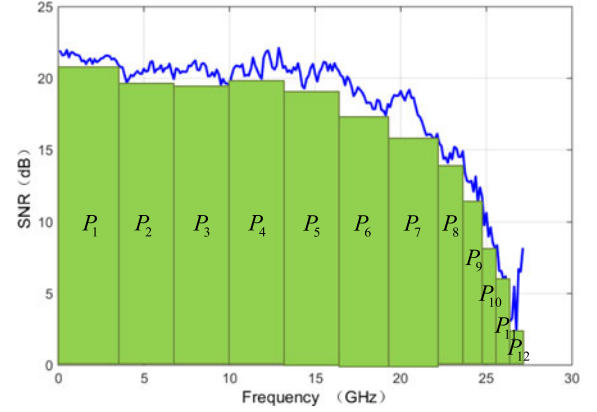


Fig. 6. Bit- or power-loading for DMT transmission system. The blue curve is the estimated SNR as a function of frequency and different amount of signal powers (represented by the height of the green rectangles) and modulation format can be encoded in each sub-carrier.

shaping and matched filters in CAP system can be implemented using analog techniques [15], we focus on digital implementation since it is the preferred choice given today's DAC and ADC technologies [8].

The DSP structure of CAP is also shown in Fig. 4. At the transmitter side, the in-phase (I) and quadrature (Q) components of the QAM signal are sent into two shaping filters with impulse responses $h_I(t) = g(t) \cdot \cos(2\pi f_c t)$ and $h_Q(t) = g(t) \cdot \sin(2\pi f_c t)$ that form a Hilbert transform pair. A square-root-raised-cosine shaping filter $g(t)$ with a roll-off coefficient of α is used as the baseband impulse response. The center frequency f_c is given by $(1 + \alpha) \cdot B/2 + \Delta f$ where Δf is the frequency offset in order to avoid spectrum aliasing. Fig. 5(a) shows the time domain response of the shaping filter. Fig. 5(b) depicts the baseband QAM signal. After the shaping filter, the electrical spectrum of generated CAP signal is shown in Fig. 5(c). The signal is shaped to be a Nyquist-like signal and up-converted to an intermediate frequency (IF) of f_c . After the shaping filter, pre-emphasis is employed to compensate for the bandwidth limitation of DAC and transmitter laser non-linearity with a time domain FIR filter. The processed signal is then fed into a DAC to generate the electrical drive signal. Fig. 5(d) shows the optical spectrum of optical CAP signal. The optical signal after transmission would be directly detected by PIN PD or APD, sampled and sent into two matched filters. The two matched filters are the time-reversed version of the shaping filters at the transmitter i.e., $m_I(t) = g(-t) \cdot \cos(2\pi f_c t)$ and $m_Q(t) = g(-t) \cdot \sin(2\pi f_c t)$ in order to separate the I and Q components. The separated I and Q signals are then combined to re-construct the complex QAM signal in digital domain which is followed by a re-sampling function. As for complex high-order QAM signals, common adaptive equalization algorithms such as constant multi modulus algorithm (CMA) [16], multi constant multi modulus algorithm (MCMA) [17] or decision directed least-mean-square (DD-LMS) can be used to compensate various channel impairments (DD-LMS will be discussed in more detail in the next section).

DMT: Also known as direct detected orthogonal frequency division multiplexing (OFDM), DMT is another attractive

modulation scheme for low cost short-reach systems [18]–[20]. In contrast with OFDM, only half of the subcarriers in DMT are coded with information while the other half are the complex conjugate of the first half to ensure that the DMT signal is real for intensity modulation in IM/DD systems. Like all multi-carrier systems, DMT enables bit loading or power loading [21] that allow a flexible set of modulation formats for each subcarrier optimized with respect to the channel transfer function to maximize overall bit rate or power margin. DMT has advantages such as high spectral efficiency, high tolerance to impairments and flexible coding. The most popular bit loading method is Chow's algorithm [22] and Fig. 6 shows the principle of bit loading. Before signal transmission, a probe signal would be transmitted and received to estimate the end-to-end SNR at each frequency sub-carrier as shown by the blue curve. In this case, high order modulation formats such as 128-QAM, 64-QAM can be coded in the subcarriers at low frequency which have high SNR. 32-QAM, 16-QAM and 8-QAM can be coded in the subcarriers at the mid-frequencies and QPSK or BPSK is coded in the subcarriers at high frequency part to achieve similar bit error ratio performance across all subcarriers. In this case, the bandwidth usage is maximized.

The DSP configuration for DMT systems is also shown in Fig. 4. After FEC encoding, the S/P function transfers the serial information bits to parallel blocks. Information bits are then mapped to QAM symbols with different modulation orders for each sub-carrier, followed by inverse fast Fourier transform (IFFT) to transform the signal to time-domain. Cyclic prefix (CP) is added to avoid inter-block interference and allow simple one tap frequency domain equalization. Pre-emphasis is added to compensate the bandwidth limitation of DAC while transmitter laser nonlinearity effects are mitigated with a time-domain FIR filter. At the receiver side, the photo-detected signal is re-sampled and undergo symbol frame synchronization, serial-to-parallel conversion, CP removal and fast Fourier transform (FFT) to obtain the data symbols on each subcarrier. A frequency-domain equalizer is used to compensate for channel distortions followed by FEC decoding and symbol decisions.

In terms of experiments, R. Rodes *et al.* demonstrated the first single-wavelength 100 Gbit/s transmission using PDM-PAM4 in 2012 [23]. K. Zhong *et al.* demonstrated single-channel PAM-4 signal transmission with bit rate up to 140 Gbit/s and 4-lane 500 Gbit/s PAM-4 transmission system using commercially available 25 GHz EML-TOSA and direct detection [14], [24]. Most recently, S. Kanazawa *et al.* demonstrated a record high 107 GBaud PAM-4 IM/DD system using an ultra-broadband EML [25]. CAP-16 and other high-order CAP signals were first demonstrated for short-reach applications by Tao *et al.* [26], [27]. Employing multi-band techniques, single lane 102 Gbit/s and four-lane 400 Gbit/s multi-band CAP transmissions were also demonstrated [28], [29]. Another related format called half-cycle M-QAM Nyquist subcarrier modulation (SCM) also attracts much interest [30]–[32] and four-lane 608 Gbit/s using half-cycle 16QAM Nyquist-SCM signal using 25 GHz EML and direct detection [33] were demonstrated. For DMT signals, single-channel 101 Gb/s DMT signal transmission over 10 km was achieved with a 64 GSa/s arbitrary

TABLE II
RECENT EXPERIMENTAL DEMONSTRATIONS OF SHORT-REACH SYSTEMS USING PAM-4, CAP-16 AND DMT FORMATS

Modulation format	Bit-Rate (Gb/s)	Number of wavelength	Distance (km)	Reference
PAM4	100	1	1	[23]
PAM4	140	1	20	[14]
PAM4	500	4	2	[24]
PAM4	214	1	2	[25]
CAP16	102	1	10	[28]
CAP16	400	4	20	[29]
Half-cycle SCM-16QAM	608	4	10	[33]
DMT	100	1	10	[34]
DMT	468	4	40	[35]
DMT	560	4	2	[36]

TABLE III
KEY PARAMETERS FOR 100 GB/s PAM-4, CAP AND DMT SHORT-REACH SYSTEMS

Modulation Format	Baud Rate	Roll off factor	Signal Bandwidth	No. of Subcarriers
PAM-4	56G	0.12	31.36 GHz	1
CAP(16QAM)	28G	0.06	29.78 GHz	1
DMT(QPSK to 64QAM)	NA	NA	31.5 GHz	256

waveform generator (AWG) and a directly modulated laser [34]. More recently, 4×117 Gb/s DMT signals have been successfully transmitted over 40 km of SSMF [35]. A four-channel 560 Gbit/s 128 QAM-DD-OFDM transmission over 2 km of SSMF was also demonstrated [36]. Table II summarizes recent demonstrations of PAM-4, CAP-16 and DMT short-reach systems.

Performance comparison of modulation formats: Each modulation format has its own advantages and drawbacks. We hereby focus on 100 Gb/s/lambda short-reach links. In [37], Zhong *et al.* compared the performance of these three modulation formats with typical commercial devices. Key parameters for different modulation formats are shown in Table III. For PAM-4 signals, the baud rate is 56 Gbaud which gives a bit rate of 112 Gbit/s. The bandwidth is 31.36 GHz using a raised cosine pulse shaping with a roll off factor of 0.12. For CAP-16, the baud rate is 28 Gbaud and the root raised cosine pulse for both in-phase and quadrature components has a roll off factor of 0.06. It should be noted that roll off factor for pulse shaping was obtained by a parameter sweeping in order to achieve best performance for each modulation. In addition, CAP-16 has a slightly smaller roll-off factor than PAM-4 since a frequency offset of 0.1GHz is used for CAP-16 here which would require the spectrum of CAP-16 to be slightly tighter for the same bandwidth. For DMT signal, FFT size is 512 with Hermitian symmetry, CP is 8, and 217 subcarriers are coded with signals from QPSK to 64QAM.

Fig. 7 shows the simulation results for single wavelength 112 Gbit/s IM/DD systems. Detailed information of simulation setup can be found in [37]. Fig. 7(a) shows the BER as a function of received optical power for different formats. FFE adapted by DD-LMS are used for PAM-4 signals at the receiver.

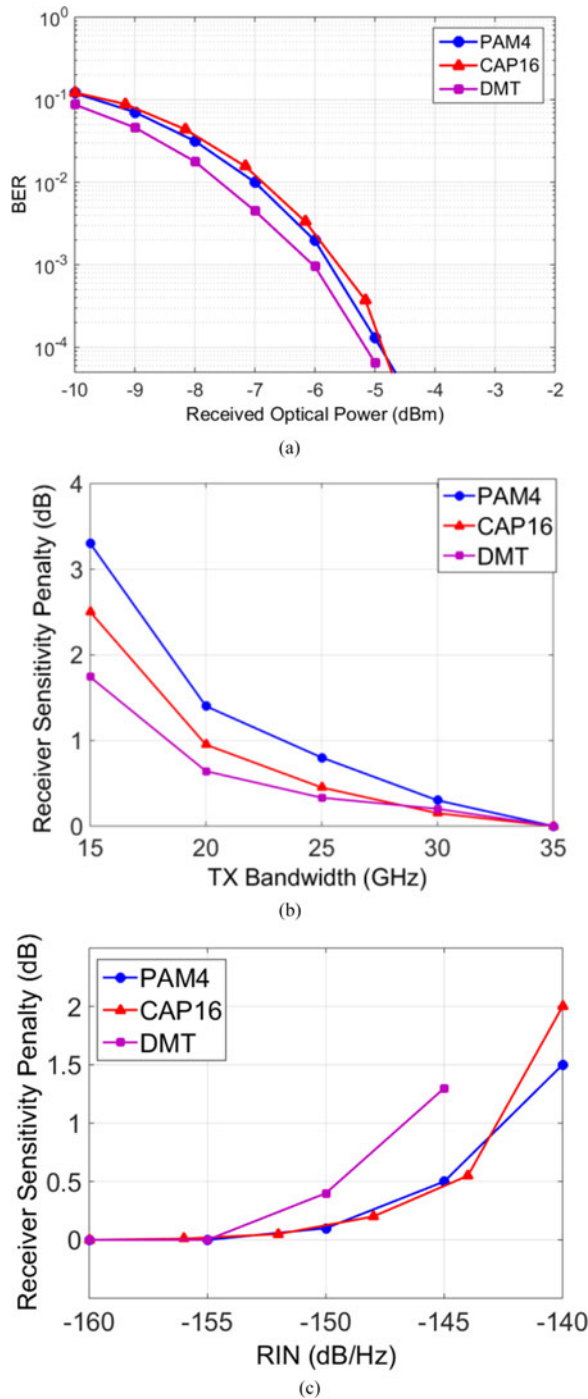


Fig. 7. Simulation results for single wavelength 112 Gbit/s IM/DD system with different modulation formats: (a) BER vs. received optical power, (b) RS penalty vs. TX bandwidth, (c) RS penalty vs. RIN [37].

The performance of DMT is the best among the three formats. PAM-4 is slightly better than CAP-16. Receiver sensitivities of -6.95 dBm, -6.4 dBm and -6.15 dBm at 7% FEC overhead limit of 3.8×10^{-3} are demonstrated for DMT, PAM-4 and CAP-16, respectively. Fig. 7(b) shows the RS penalty as a function of transmitter bandwidth. The received optical power at BER = 3.8×10^{-3} for transmitter bandwidth of 35 GHz was used as the reference. It can be seen that PAM-4 is most sensitive to

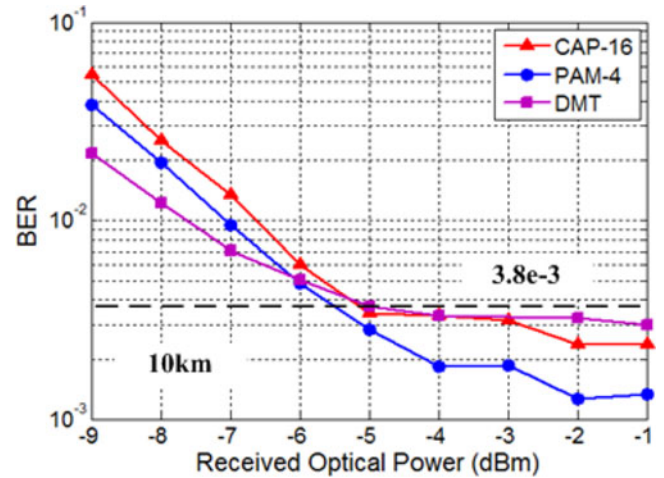


Fig. 8. Experimental results for single wavelength 112 Gbit/s IM/DD system with different modulation formats for 10 km transmission. [37]

transmitter bandwidth. On the other hand, CAP-16 occupies less bandwidth with respect to PAM-4 for the same data rate due to high spectral efficiency and sharp pulse shaping and hence is more tolerant to transmitter bandwidth. DMT provides the best transmission performance because of the availability of bit loading. Fig. 7(c) shows the RS penalty as a function of RIN for different formats. PAM-4 and CAP-16 have similar tolerance to RIN. Moreover, as DMT waveforms resemble an analog signal with higher peak-to-average power ratio (PAPR), they are more sensitive to RIN, ADC/DAC resolution among others. At 1 dB penalty, the tolerances to RIN are -147 dB/Hz, -143 dB/Hz and -143 dB/Hz for DMT, PAM-4 and CAP-16, respectively.

Fig. 8 shows the experimentally measured BER as a function of received optical power for 112 Gbit/s IM/DD transmission over 10 km with different modulation formats using 25 Gbps device. It can be seen that PAM-4 slightly outperforms CAP-16. At received optical power regime above -5 dBm in which the BER is below the HD-FEC threshold, the performance of DMT is the worst among the three modulation formats and a BER floor of 3×10^{-3} was observed for the DMT signal. This is due to the high PAPR of DMT signal, limited 8-bit resolution of DAC/ADC and the nonlinearity of the driver amplifier for EML. Receiver sensitivities of -5.2 dBm, -5.1 dBm and -5.6 dBm were obtained for CAP-16, DMT and PAM-4, respectively.

Complexity comparison of modulation formats: The required hardware (driver, transmitter and receiver etc.) for PAM-4, CAP-16 and DMT is similar. Instead, the biggest difference between the modulation formats lies in the transmitter and receiver DSP. The DSP complexity (including DACs and ADCs) is a critical factor that affects cost, power consumption and hence practical implementation. While all three formats share a few common DSP blocks, we will highlight the major differences. Firstly, PAM-4 is the simplest among the three formats in terms of signal generation. Pulse shaping is an option to reduce the signal bandwidth but not necessary. For demodulation, only threshold decision is required for PAM-4 signal. However, for CAP-16, a pair of shaping filters is required to combine I and Q components of complex QAM signal to generate a real CAP-16

TABLE IV
COMPARISON OF DIFFERENT MODULATION FORMATS

Modulation Format	RS	RIN	BW	ADC /DAC	EQ	DSP Complexity
PAM-4	fair	good	bad	good	fair	fair
CAP-16	fair	good	fair	fair	fair	fair
DMT	good	bad	good	bad	good	good

BW: bandwidth, EQ: equalization

signal. At the receiver side, a pair of matched filters is required to separate I and Q components from the received CAP-16 signal. The shaping and match filters can be realized by digital FIR. But to achieve more spectrally efficient rectangle spectral shaping, long FIR with large number of taps is needed. DMT, which encodes information on frequency subcarriers, requires IFFT function for DMT signal generation at transmitter and FFT function for demodulation at receiver. In addition, bit loading and power loading are required for system optimization which also increase the DSP complexity. For signal generation and demodulation, PAM-4 is the simplest, DMT requires most DSP resource and CAP-16 lies in between. Moreover, the equalization complexity of the three different modulation formats are compared in terms of real multipliers per bit. 301 and 43 real multipliers per bit are required for PAM-4 with time-domain equalization (TDE) and frequency-domain equalization (FDE) respectively. 198(TDE) and 40(FDE) real multipliers per bit are required for CAP-16. For DMT, only 21 real multipliers per bit are required. Therefore, the computational complexities of PAM-4 and CAP-16 are similar while DMT is simpler as it only needs one tap equalizer. More in-depth discussions about the computational complexity can be found in [37]. Last but not least, the ADC/DAC resolution requirements are also compared. In theory, a resolution of 2 bits is sufficient for PAM-4 while DMT needs much higher resolution due to its analog-like nature and the difficulty to fully suppress PAPR effects [38]–[40]. DMT also requires good linear electrical amplifiers. In this regard, CAP-16 is in between PAM-4 and DMT and is less demanding on ADC/DAC than DMT but more than PAM-4.

Table IV shows the comparison of different modulation formats in terms of RS, RIN, TN, requirement on DAC/ADC, complexity of equalization and complexity of signal generation and demodulation. Taken into account various performance attributes, we believe that PAM-4 is the most desirable choice for high speed short-reach transmission systems as it is relatively simple to implement with good performance. The Industry is also focusing on PAM-4 as new optical modules and chipsets based on PAM-4 are being produced in recent years as PAM-4 has been chosen as a standard for 400 G Ethernet [9]. Meanwhile, the performance of CAP is good but it does not demonstrate unique advantages that outperform other formats in practical application scenarios. DMT performs best in terms of RS, bandwidth tolerance, thermal noise and complexity of equalization. However, complexity of generation and demodulation, bit loading, high requirements on ADC/DAC and linearity of transmitter and receiver, poor tolerance on RIN standardization difficulties are issues to be solved for practical implementation.

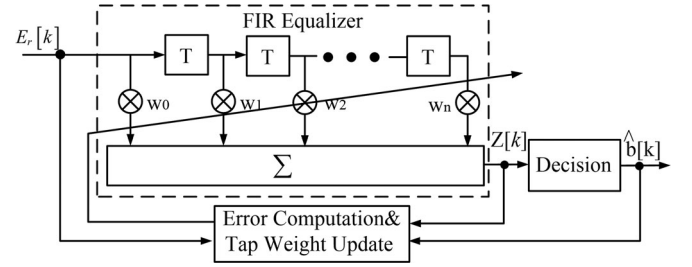


Fig. 9. Block diagram of the FFE.

Finally, comparisons of higher order PAM and CAP modulation formats can be found in [10].

IV. DIGITAL SIGNAL PROCESSING FOR SHORT REACH SYSTEMS

In short reach links, the most important transmission impairments are bandwidth limitations of transmitter and receiver, nonlinearity of EML/DML and distortions due to interaction with CD, real signaling and direct detection as the transmission distance increase. We have briefly mentioned a few key DSP blocks such as FFE as equalization techniques in the last section. In this section, we describe in detail the DSP algorithms to compensate these impairments.

FFE: Feed-forward equalizer is a widely used equalizer for linear impairments compensation. Fig. 9 shows the block diagram of FFE. The FFE output is expressed as

$$z[k] = \sum_{i=0}^{N-1} w_i E_r((k-i)T) = \mathbf{w} \cdot \mathbf{E}_r^T[k] \quad (6)$$

where $z[k]$ is the equalizer output, $\mathbf{w} = [w_0 w_1 w_2 \dots w_{N-1}]$ is the vector of tap weights, N is the number of taps and $\mathbf{E}_r[k] = [E_r(kT) E_r((k-1)T) \dots E_r((k-N+1)T)]$. The tap weights can be adaptively updated by many different strategies and interested readers are referred to [41], [42] for a more in-depth treatment of adaptive DSP for optical communications. Here, we will focus on one of the most common adaptive signal processing technique – the DD-LMS algorithm. In this case, we define the error $\varepsilon[k] = \hat{b}[k] - z[k]$ between $z[k]$ and the decided symbol $\hat{b}[k]$, the cost function $J(\mathbf{w}) = (\varepsilon[k])^2 = (\hat{b}[k] - \mathbf{w} \cdot \mathbf{E}_r^T[k])^2$ and use the iterative method of stochastic gradient descent [42] to obtain the best filter configuration that minimize $J(\mathbf{w})$. If we start our iteration with the signals at time $k_0 T$, the $n+1$ th update of the filter tap weights is given by

$$\begin{aligned} \mathbf{w}^{(n+1)} &= \mathbf{w}^{(n)} + \mu \left. \frac{\partial J}{\partial \mathbf{w}} \right|_{\mathbf{w}^{(n)}} \\ &= \mathbf{w}^{(n)} + \mu \varepsilon[k_0 + n] \mathbf{E}_r[k_0 + n] \end{aligned} \quad (7)$$

where μ is the step size. It should be noted that FFE can operate at symbol rate sampling or higher. For symbol rate FFE, a filter matched to the channel is required prior to sampling and filtering. If FFE is operated at 2 samples per symbol, the FFE can be the matched filter itself. In this case, T equals to $T_s/2$ while the tap weights are only updated at time slots $2T, 4T, 6T$ etc.

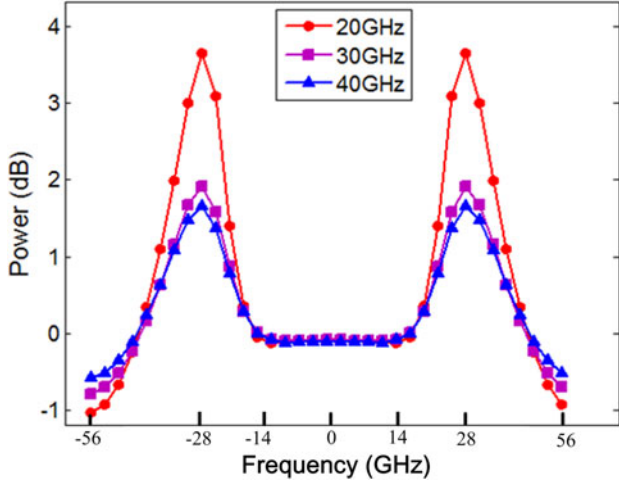


Fig. 10. Frequency response of FFE for 56 Gbaud PAM-4 system with different transceiver component bandwidths.

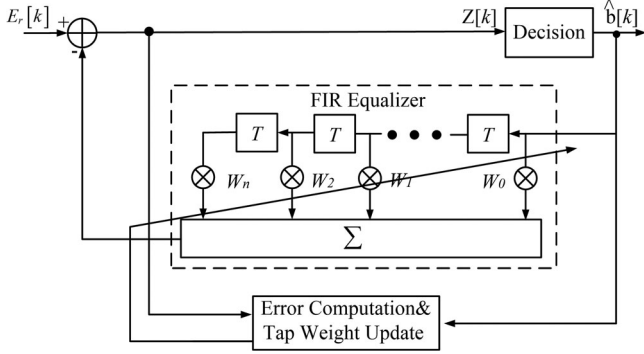


Fig. 11. Block diagram of the DFE.

We simulate a 56 Gbaud PAM-4 system with various amount of low-pass filtering using 4th order Bessel filters and study the corresponding FFE characteristics. Fig. 10 shows the frequency response of FFE for 56 Gbaud PAM4 system with different bandwidths. The number of taps is 51 for FFE. It can be seen that FFE boosts the power of high frequency components that undergo large losses due to the system bandwidth limitations.

DFE: Decision feedback equalizer is an alternative equalizer for channel impairments. Fig. 11 shows the block diagram of DFE. Different from FFE, the input of DFE are the symbols after decision $\hat{b}[k]$ as shown in Fig. 11. DFE is usually operated at 1 sample per symbol. In this case, the signal compensated by the DFE output before decision is given by

$$z[k] = E_r(kT) - \sum_{i=0}^{N-1} w_i \hat{b}[k-i] \quad (8)$$

If we start our DD-LMS iteration with the signals at time $k_0 T$, the $n+1$ th updates for the tap weights are given by

$$\mathbf{w}^{(n+1)} = \mathbf{w}^{(n)} + \mu \varepsilon[k_0 + n] \mathbf{Z}[k_0 + n] \quad (9)$$

where $\mathbf{Z}[n] = [z[n] z[n-1] \dots z[n-(N-1)]]$ and $\varepsilon[n] = \hat{b}[n] - z[n]$. It should be noted that FFE is simple and does not suffer from feedback delays but is ineffective in equalizing spectral nulls. DFE on the other hand can successfully equalize

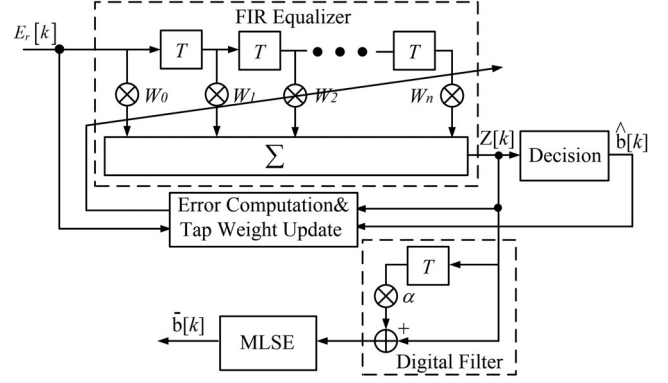


Fig. 12. Block diagram of DD-FTN.

spectral nulls by pole insertion but it may become unstable and suffer from decision error propagation. Nonetheless, since the CD-induced channel impulse response contain both pre-cursor (future symbols) and post-cursor (past symbols) ISI, the best choice for equalization is a combination of a feed-forward and a feedback structure equalizer. Recently, Tomlinson-Harashima pre-coding, a transmitter-side DFE that does not suffer from error propagation, is shown to outperform DFE in various PAM-4 short-reach transmissions scenarios [43].

Direct detection-faster than nyquist (DD-FTN): While FFE and DFE are widely used, it is well known that they severely enhance the in-band noise in bandwidth-limited optical transmission systems. This can be easily illustrated by the frequency response of FFE shown in Fig. 10. As the equalizer tries to boost the power of high frequency components to compensate for the low-pass effects, it also amplifies the noise power at high frequency components. Noise enhancement results in a SNR degradation of the equalized signal. To address such problems, we proposed a DD-FTN algorithm [14] and the schematic diagram is shown in Fig. 12.

The DD-FTN consists of a FFE equalizer adapted by LMS, a digital post filter and maximum likelihood sequence estimation (MLSE) function. The post filter is placed after the equalizer for in-band noise suppression. The response of the post filter has a transfer function $H(z) = 1 + \alpha z^{-1}$. The tap coefficient α is introduced to facilitate the optimization of frequency response of the post filter. Fig. 13(a) shows the frequency response of post filter for different tap coefficients. It can be seen that larger tap coefficients result in lower bandwidth, sharper roll-off and larger attenuation at high frequencies. Normally, α increases with signal baud rate/bandwidth. The post filter suppresses the noise but induces ISI, which is further addressed by MLSE. Although DD-FTN includes two more DSP blocks compared to conventional equalizers, the complexity increments are quite small. The post filter is only a delay and addition. As studied in part E of [44], MLSE requires $3M^2$ additions, M^2 Multiplications, and $M(M-1)$ comparisons for M -ary PAM signals. However, approaches such as state grouping is proposed to merge several states for trellises originally with a large number of states [45], [46]. In this case, the state count can be halved to merely two for PAM-4 signal, which significantly reduces the complexity. In the meantime, we have found that a mem-

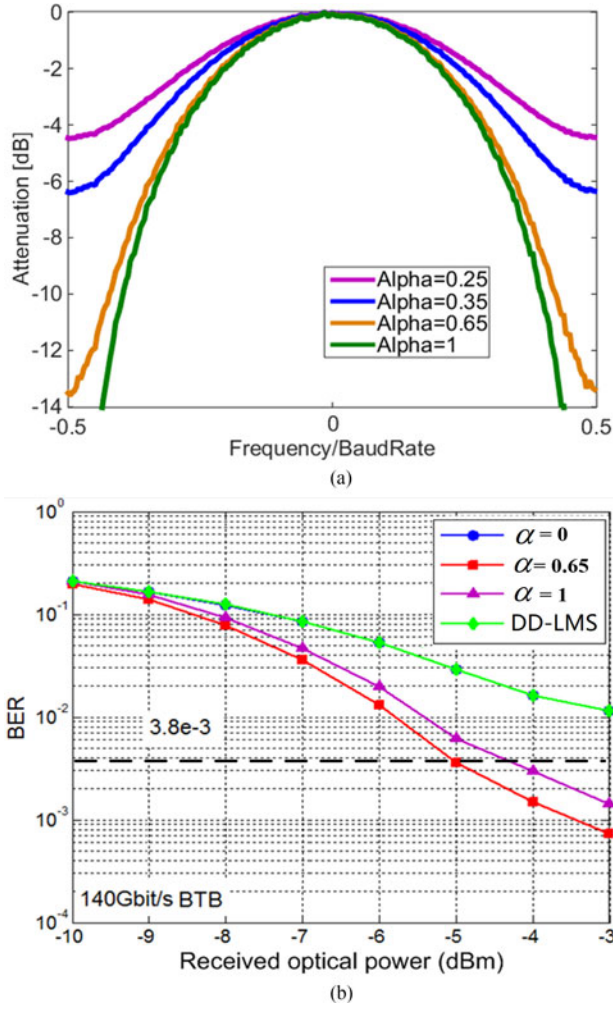


Fig. 13. (a) Frequency response of post filter for different α . (b) BER vs. received optical power (ROP) for 70 GBaud PAM-4 IM/DD system with DD-FTN [14].

ory length of 20 symbols is sufficient to ensure good detection performance. Therefore, the overall complexity of DD-FTN is not high compared to other DSP approaches.

Fig. 13(b) shows the BER vs. received optical power for a 140 Gbit/s PAM-4 IM/DD transmission experiment with and without DD-FTN [14]. The BER with DD-LMS only is also shown (green curve). With a tap coefficient of $\alpha = 0$ (blue curve), the performance is the same with that of DD-LMS only (hence the blue and green curve overlap with each other), which exhibits the worst performance. For received optical power of -4 dBm, a BER of 1.7×10^{-2} is obtained. BER decreases significantly when tap coefficients $\alpha = 1$ and $\alpha = 0.65$ are used. For a received optical power of -4 dBm, BER of 3×10^{-3} and 1.6×10^{-3} are obtained for $\alpha = 1$ and $\alpha = 0.65$, respectively. The best performance is obtained with optimal tap coefficient $\alpha = 0.65$. Receiver sensitivities at the 7% FEC limit of 3.8×10^{-3} are -5 dBm and -4.3 dBm for the cases of $\alpha = 0.65$ and $\alpha = 1$, respectively. Transmission over 20 km at 1296.2 nm is experimentally demonstrated.

Volterra series based nonlinear equalizer (VNLE): While linear distortions can be efficiently compensated using FFE, DFE

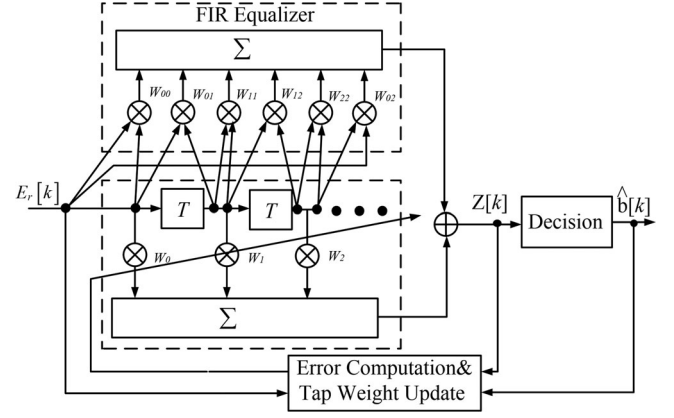


Fig. 14. Block diagram of the VNLE.

or DD-FTN, nonlinear distortions such as modulation chirp, saturated power amplification, nonlinear modulation characteristics of lasers or modulators and square law detection for the signal with CD [47]–[49] are difficult to compensate effectively.

One of the most common algorithms to compensate nonlinear effects is VNLE. Since most of the nonlinear impairments in IM/DD transmission system such as saturated power amplification, nonlinearity from modulator and CD induced ISI etc are primarily second-order nonlinearities, second order VNLE is introduced in this paper. It should be noted that VNLE can be implemented based on FFE or DFE. Here, we focus on the VNLE based on FFE. Fig. 14 shows the schematic diagram of VNLE with input $E_r[k]$ and output

$$\begin{aligned}
 Z[k] &= Z_L[k] + Z_{NL}[k] \\
 &= \sum_{i=0}^{N_L-1} w_{L,i} E_r((k-i)T) \\
 &\quad + \sum_{i=0}^{N_{NL}-1} \sum_{j=0}^{N_{NL}-1} w_{NL,ij} E_r((k-i)T) E_r((k-j)T)
 \end{aligned} \tag{10}$$

Where $Z_L[k]$ and $Z_{NL}[k]$ is the linear and nonlinear equalized output respectively, $w_{L,i}$ is the tap weight for linear part of VNLE, $w_{NL,ij}$ is the tap weight for nonlinear part of VNLE, N_L is the number of taps for the linear part of VNLE and N_{NL} is the number of taps of nonlinear part of VNLE. Representing the filter taps as vectors $\mathbf{w}_L$ and matrix $\mathbf{w}_{NL}$ and if we start our DD-LMS iteration with the signals at time $k_0 T$, the $n+1^{th}$ tap weights update will be given by

$$\mathbf{w}_L^{(n+1)} = \mathbf{w}_L^{(n)} + \mu \varepsilon[k_0 + n] \mathbf{E}_r[k_0 + n] \tag{11}$$

$$\mathbf{w}_{NL}^{(n+1)} = \mathbf{w}_{NL}^{(n)} + \mu \varepsilon[k_0 + n] \mathbf{E}_r[k_0 + n] (\mathbf{E}_r[k_0 + n])^T \tag{12}$$

where $\varepsilon[n] = \hat{b}[n] - z[n]$. Zhang *et al.* employed VNLE in an SSB-DMT system [47] and an average SNR improvement of 2 dB across subcarriers for a 100 G SSB-DMT signal is obtained by using VNLE. Fig. 15 shows the measured BER vs. OSNR with and without VNLE, clearly indicating that VNLE

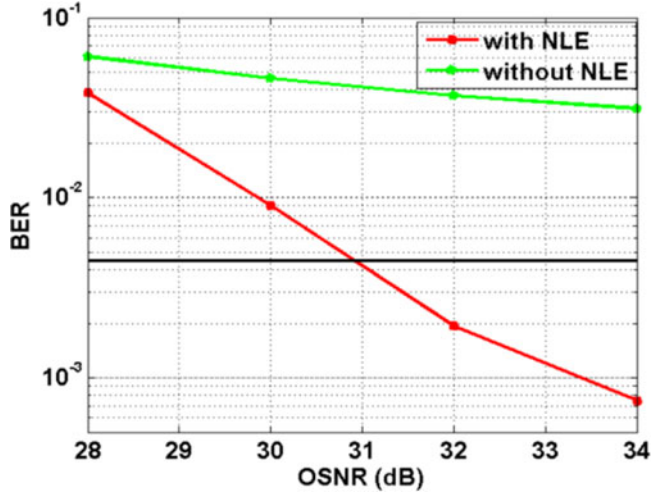


Fig. 15. BER vs. OSNR with and without VNLE for 100 Gb/s single-side band-discrete multi-tone (SSB-DMT) transmission experiments [47].

significantly improves system BER performance. It should be noted that for systems with high carrier to signal power ratio (CSPR), the signal component is smaller and hence the overall system response will be more linear, rendering VNLE less effective.

SSB signaling: For inter-data center interconnection applications, the transmission distance can be as large as 80 km. Due to large attenuation of the O-band (0.35 dB/km) and lack of good O-band amplifiers, C-band with low attenuation and mature erbium-doped fiber amplifier (EDFA) technology is preferred. Furthermore, the current two WDM standards in O-band are CWDM4 (1271 nm 1291 nm 1310 nm 1330 nm) and LAN-WDM (1295.56 nm, 1300.05 nm, 1304.58 nm and 1309.14 nm) and commercially available WDM MUX in O-band can only support 4 channels in accordance to the standards. Therefore, these factors limit the potential of substantial capacity upgrades for long-distance inter-connect applications in the near future. However, the main challenge at C-band for IM/DD systems up to 80 km is the channel fading induced by CD after direct detection. Liu *et al.* used DMT with bit loading and power loading to combat such fading and experimentally demonstrated 49 Gbit/s DMT signal transmission over 80 km [50]. However, it is difficult to further increase the system data rate. Dispersion compensation fiber (DCF) can be installed in the transmission system to compensate fiber CD. However, they are generally bulky, induce high signal loss and increase system cost. CD pre-compensation using complex IQ or dual-drive Mach-Zehnder modulator (DDMZM) modulator is another promising way to combat CD effects [51]–[55]. However, it is difficult to have prior knowledge of CD for the transmission links in an elastic optical network (EON). In addition, high PAPR due to CD pre-compensation will result in performance degradation. Non-zero dispersion shifted fiber (NZ-DSF) is a good alternative in principle as they have much lower CD compared to SSMF. However, SSMF dominates current research activities and deployments due to its low cost and interest in NZ-DSF has yet to grow.

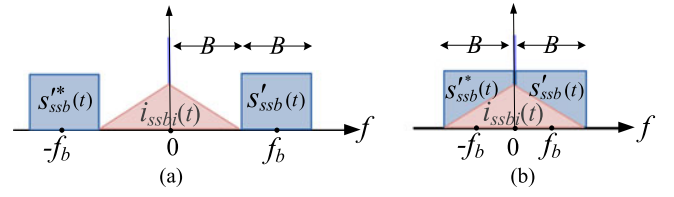


Fig. 16. Electrical spectrum of received SSB signal. (a) $f_b = 1.5 B$. (b) $f_b = 0.5 B$.

In light of the above constraints, SSB signaling becomes a promising approach to combat CD effects [56]–[65]. A transmitted optical SSB signal can be expressed as

$$E_t(t) = A + s_{ssb}(t) e^{j2\pi f_b t} \quad (13)$$

where A is the direct current bias and $s_{ssb}(t) e^{j2\pi f_b t}$ is the SSB signal at frequency f_b . After propagation, the detected signal can be written as

$$E_r(t) = |E_t(t) \otimes h(t)|^2 \quad (14)$$

where $h(t)$ is the channel response of the optical link including bandwidth limitation of components and CD. Noise is not included here for simplicity. Substituting (13) into (14), the detected signal $E_r(t)$ can be expressed as

$$\begin{aligned} E_r(t) &= |(A + s_{ssb}(t) e^{j2\pi f_b t} \otimes h(t))|^2 \\ &= |A|^2 + 2\text{Re}\{s'_{ssb}(t)\} + i_{ssbi}(t) \end{aligned} \quad (15)$$

where $s'_{ssb}(t) = A \cdot s_{ssb}(t) e^{j2\pi f_b t} \otimes h(t)$ and $i_{ssbi}(t) = |s_{ssb}(t) e^{j2\pi f_b t} \otimes h(t)|^2$. The first term of (15) is a direct current component. The second term is the desired received signal and the last term is the SSB-induced second order nonlinear signal-to-signal beating interference (SSBI). Assuming that the side-band signal has bandwidth B , the unwanted SSBI will occupy a frequency band from $-B$ to $+B$ as shown in Fig. 16. Therefore, the frequency f_b is an important parameter to determine the type of SSB system and corresponding DSP approaches.

There are two scenarios for SSBI mitigation. In the first scenario, there is a guard band for the SSB signal so that the SSBI term do not overlap with the desired signal as shown in Fig. 16(a). Here, f_b is required to be larger than $1.5 B$ to prevent the desired signal from being interfered by SSBI at the cost of spectral efficiency. In this case, the desired signal can be extracted and further processed through standard filtering techniques. In the second scenario with zero guard-band (ZG) between the SSB signal and the carrier, the desired signal and SSBI overlap in frequency and additional SSBI cancellation techniques has to be employed.

For ZG-SSB systems with OFDM signaling, one can use iterative methods to mitigate SSBI and its operating principle is depicted in Fig. 17. The upper and lower branches are responsible for channel compensation and SSBI reconstruction respectively. After direct current removal and OFDM window synchronization, the detected SSB-OFDM signal $E'_r(t) = 2\text{Re}\{s'_{ssb}(t)\} + i_{ssbi}(t)$ is filtered to only retain the positive frequency spectrum and converted into frequency

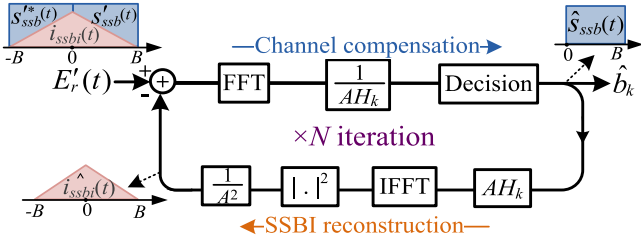


Fig. 17. Principle of SSBI elimination through iteration method. SF: Sideband filtering.

domain. Here, the k th received subcarrier can be expressed as

$$C_k = AH_k b_k + I_k \quad (16)$$

where b_k , H_k and I_k are the transmitted information symbols, channel response, and SSBI product at the k th subcarrier, respectively. Accurate estimation of channel transfer function H_k plays an important role in the stability of the whole iterative SSBI cancellation system. In the presence of SSBI, AH_k can be estimated by using frequency interleaved training symbols [60], [61] (only odd subcarriers are filled) or an orthogonal pair of training symbols [62]. The estimated transfer function is used to equalize the received signal and the equalizer output of k th subcarrier is given by

$$C'_k = b_k + \frac{1}{A} I_k H_k^{-1} \quad (17)$$

and the information symbol estimate $\hat{b}_k$ can be obtained by symbol decision on C'_k . It can be seen from (17) that the probability of symbol decision error depends on the carrier amplitude A . For a small CSPR, the SSBI interference is relatively large and thus the equalized signal suffers dominant effect from SSBI rather than noise. For a large CSPR, the interference can be suppressed effectively. Therefore, to ensure correct iteration convergence process and avoid error propagation, a larger CSPR is required for ZG-SSB systems compared to SSB systems with guard band.

The estimated information symbol $\hat{b}_k$ will be fed back into the lower branch of Fig. 17 with de-equalization and IFFT operation to re-construct an SSBI estimate

$$\begin{aligned} \hat{i}_{ssbi}(t) &= \left| \sum_{k=0}^N \hat{b}_k \cdot \hat{H}_k e^{j2\pi f_k t} \cdot e^{j2\pi f_b t} \right|^2 \\ &= |\hat{s}_{ssb}(t) \cdot e^{j2\pi f_b t} \otimes h(t)|^2 \end{aligned} \quad (18)$$

and will be subtracted from the original received signal $E_r'(t)$. To obtain optimum SSBI cancellation performance, more than 5 iterations are typically required in practice. This results in higher computational complexity, latency and larger system memory than SSB systems with guard band. It should be noted that since CSPR, channel response, receiver responsivity and other component parameters are unknown to the receiver in practice, the estimated channel response and carrier power in (17) will not be exactly AH_k and A respectively. An additional scaling factor will be inherently captured by the training symbols which will ensure detection accuracy.

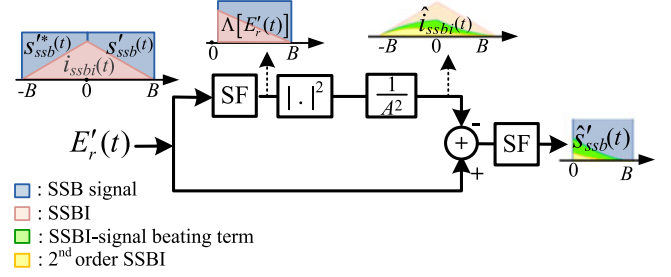


Fig. 18. Operating principle of feed-forward SSBI cancellation [63]. SF: Sideband filter.

Moreover, Randel *et al.* proposed an alternative SSBI cancellation approach with simpler single-stage feed-forward mode as shown in Fig. 18 [63]. Compared to the aforementioned iterative SSBI cancellation process [64], this method is more advantageous as FIR-based adaptive equalization is computationally simple, induce minimal latency and does not require a lot of system memory. In particular, after direct current component removal and timing recovery, the detected signal $E_r'(t) = 2\text{Re}\{s'_{ssb}(t)\} + i_{ssbi}(t)$ is filtered using a digital sideband filter(SF) $\Lambda\{\cdot\}$ followed by a square-law operation to reconstruct the SSBI product as

$$\begin{aligned} \hat{i}_{ssbi}(t) &= |\Lambda\{E_r'(t)\}|^2/A^2 = |s'_{ssb}(t) + \Lambda\{i_{ssbi}(t)\}|^2/A^2 \\ &= |s'_{ssb}(t)|^2/A^2 + 2\text{Re}\{\Lambda\{i_{ssbi}(t)\} \cdot s'^*_{ssb}(t)\}/A^2 \\ &\quad + |\Lambda\{i_{ssbi}(t)\}|^2/A^2 \\ &= i_{ssbi}(t) + \underbrace{2\text{Re}\{\Lambda\{i_{ssbi}(t)\} \cdot s'^*_{ssb}(t)\}/A^2}_{\text{SSBI-Signal beating interference}} \\ &\quad + \underbrace{|\Lambda\{i_{ssbi}(t)\}|^2/A^2}_{\text{2nd order SSBI}}. \end{aligned} \quad (19)$$

The terms in (19) are the reconstructed SSBI, SSBI-signal beating component and 2nd order SSBI respectively as shown in Fig. 18. Note that in practice, the scaling factor is affected by transceiver component non-idealities such as insertion loss (IL) among others and will not be exactly $1/A^2$. Therefore, similar to other SSBI cancellation schemes, training symbols may be needed to estimate an appropriate scaling factor value in advance. After optimum scaling, the new SSB signal estimate $\hat{s}'_{ssb}(t)$ can be obtained by subtracting $\hat{i}_{ssbi}(t)$ from $E_r'(t)$ followed by sideband filtering

$$\begin{aligned} \hat{s}'_{ssb}(t) &= \Lambda\{E_r'(t) - \hat{i}_{ssbi}(t)\} \\ &= s'_{ssb}(t) - \underbrace{\Lambda\{2\text{Re}\{\Lambda\{i_{ssbi}(t)\} \cdot s'^*_{ssb}(t)\}/A^2\}}_{\text{SSBI-Signal beating interference}} \\ &\quad - \underbrace{\Lambda\{|\Lambda\{i_{ssbi}(t)\}|^2/A^2\}}_{\text{2nd order SSBI}} \end{aligned} \quad (20)$$

In this case, the original SSBI $i_{ssbi}(t)$ is eliminated but two new interference terms are introduced into the SSB signal. While they may assume a fairly complicated mathematical structure,

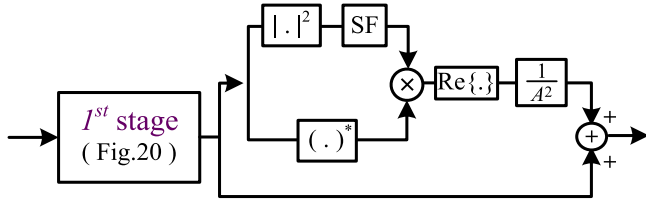


Fig. 19. Operating principle of two-stage SSBI cancellation [65]. SF: Side-band filter.

their power levels are proportional to the square and cube of CSRR. Therefore, the new interference terms are much smaller than the original SSBI $i_{ssbi}(t)$ for typical CSRR values. On the other hand, the signal OSNR is reduced when CSRR is increased. Consequently, greater interference suppression ability has to be weighed up against larger noise when considering overall performance of SSB systems.

To further mitigate the new beating interference terms of the feed-forward SSBI cancellation output $\hat{s}'_{ssb}(t)$, Li *et al.* proposed a two-stage SSBI mitigation method [65] as shown in Fig. 19 in which the single-stage feed-forward SSBI mitigation in Fig. 18 is employed in the first stage and the second stage is used to mitigate the remaining signal-SSBI beating product, i.e., $\Lambda\{2\text{Re}\{\Lambda\{i_{ssbi}(t)\} \cdot s'_{ssb}(t)^*\}/A^2\}$. Like the one-stage approach, this two-stage process will produce additional interference terms, but their magnitudes are even smaller. Finally, twin-SSB systems [70] were proposed recently to double the spectral efficiency but accurate and sharp optical filters will be required at the receivers to decompose them into two separate SSB systems for detection.

In 2016, Mecozzi *et al.* proposed a direct detection coherent receiver based on the Kramers-Kronig (KK) relations [66]. KK receiver can reconstruct optical phase information of the signal from the detected amplitude waveform, which allows one to easily compensate CD using receiver DSP. Two conditions need to be satisfied for a KK receiver: 1) optical SSB signal is required, 2) large CSRR is required to achieve minimum phase condition. Here, let $s(t)$ be a complex signal whose spectrum is contained between $-B/2$ and $B/2$. Consider an optical SSB signal

$$E_t(t) = A + s(t) \exp(-j\pi Bt) \quad (21)$$

with $i(t) = |E_r(t)|^2$ as the directly detected current. Assuming that A is large enough to ensure that $E_r(t)$ is minimum phase, the amplitude and phase information of signal can be re-constructed as

$$s(t) = \left\{ \sqrt{i(t)} \exp[j\phi_S(t)] - A \right\} \exp(j\pi Bt) \quad (22)$$

$$\phi_S(t) = \frac{1}{2\pi} p.v. \left\{ \int_{-\infty}^{\infty} dt' \frac{\log[i(t')]}{t - t'} \right\} \quad (23)$$

where *p.v.* stands for principal value. A wave of high speed transmission experiments based on KK receivers were demonstrated [67]–[72]: Zhe. Li *et al.* experimentally demonstrated 4×168 Gb/s 35 GHz-spaced WDM single-polarization direct-detection 64-QAM SSB Nyquist-SCM transmission over 80 km of uncompensated SSMF at a record net ISD of 4.54 b/s/Hz us-

ing KK receiver [67]. Le *et al.* experimentally demonstrated 240-Gb/s/λ signals over 125 km fiber [68]. Chen *et al.* experimentally demonstrated 256-Gb/s/λ signals over 160 km fiber aided with a digital carrier [69]. Zhuge *et al.* proposed a spectrally efficient and cost-effective twin-SSB system based on KK receiver optical phase reconstruction, and demonstrated 88.3 Gbit/s net bit rate transmissions with 40GSa/s DACs over 80 km SSMF at 1550 nm [70]. For PDM systems, Hoang *et al.* experimentally demonstrated 480 Gbit/s PDM-16QAM single carrier signal over 80 km SSMF with a BER below the threshold of $2E-2$ [71]. X. Chen *et al.* has successfully demonstrated 4×240 Gbit/s KK-based WDM PDM-32-QAM system over 125 km SSMF [72].

It should be noted that KK receivers require lower CSRR to satisfy minimum phase condition compared to that required for ZG-SSB systems to compensate SSBI effectively. Also, due to its inherent carrier generation, low-cost modulators such as dual-drive(DD)-MZM are naturally suited for KK systems.

V. POLARIZATION-DIVISION MULTIPLEXED TRANSMISSIONS FOR SHORT REACH SYSTEMS

PDM provides 4 degrees of freedom for encoding information and double the transmission capacity for coherent systems while standard IM/DD systems support 1 degree of freedom hence 1-dimensional modulation only. For short-reach systems limited by direct detection and other low-cost implementation constraints, innovative transceiver configurations and DSP are actively being pursued to exploit such additional degrees of freedom inherent in polarization multiplexing. One of the key research direction in this regard is SV-DD transceivers. Besides Jones space, an alternative representation to describe the SOP of an optical signal is the Stokes space representation which expresses the SOP in optical powers instead of electric field. The transmitted SV consists of four Stokes parameters

$$\begin{aligned} S_{t,0} &= |E_{t,x}|^2 + |E_{t,y}|^2, & S_{t,2} &= 2\text{Re}\{E_{t,x}E_{t,y}^*\} \\ S_{t,1} &= |E_{t,x}|^2 - |E_{t,y}|^2, & S_{t,3} &= 2\text{Im}\{E_{t,x}E_{t,y}^*\} \end{aligned} \quad (24)$$

and $S_{t,0} = \sqrt{S_{t,1}^2 + S_{t,2}^2 + S_{t,3}^2}$. With polarization rotation, the received SV is given by

$$\begin{aligned} \begin{bmatrix} S_{r,0} \\ S_{r,1} \\ S_{r,2} \\ S_{r,3} \end{bmatrix} &= \begin{bmatrix} |E_{r,x}|^2 + |E_{r,y}|^2 \\ |E_{r,x}|^2 - |E_{r,y}|^2 \\ 2\text{Re}\{E_{r,x}E_{r,y}^*\} \\ 2\text{Im}\{E_{r,x}E_{r,y}^*\} \end{bmatrix} \\ &= \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \cos \varepsilon & \sin \theta \sin \varepsilon \\ 0 & \sin \theta & \cos \theta \cos \varepsilon & -\cos \theta \sin \varepsilon \\ 0 & 0 & \sin \varepsilon & \cos \varepsilon \end{bmatrix}}_{\text{SOP Rotation matrix in Stokes space}} \cdot \begin{bmatrix} S_{t,0} \\ S_{t,1} \\ S_{t,2} \\ S_{t,3} \end{bmatrix} \\ &= \mathbf{R} \cdot \begin{bmatrix} S_{t,0} \\ S_{t,1} \\ S_{t,2} \\ S_{t,3} \end{bmatrix} \end{aligned} \quad (25)$$

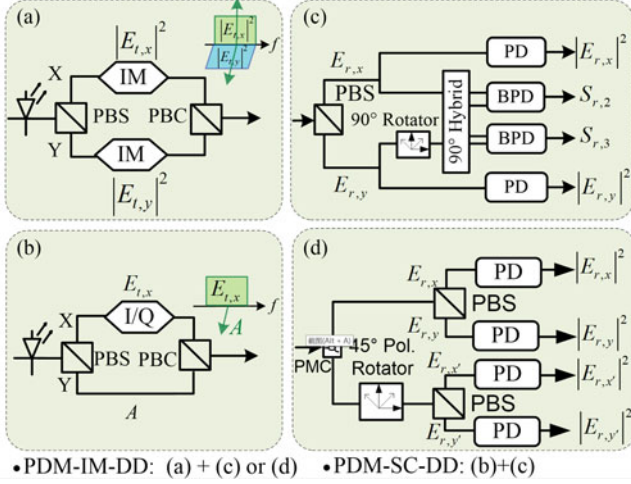


Fig. 20. Different combinations of transceiver architectures for 2D PDM-DD systems. (a) Tx: IM + IM. (b) Tx: IQ + Carrier. (c) Rx: SV-DD receiver. (d) Rx: Four PDs receiver.

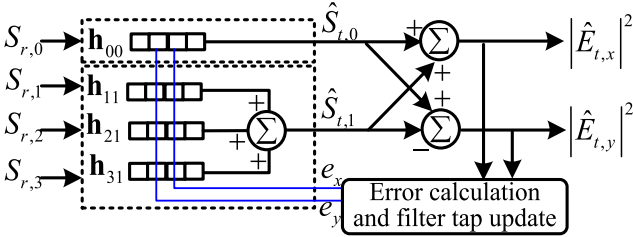


Fig. 21. Receiver DSP block diagrams for 2D PDM-IM-DD system.

where $\mathbf{R}$ is the mathematical equivalent of H_{SOP} of (4) in Stokes space. The SV includes four components, i.e., $|E_{t,x}|^2$, $|E_{t,y}|^2$, $\text{Re}\{E_{t,x}E_{t,y}^*\}$ and $\text{Im}\{E_{t,x}E_{t,y}^*\}$ which actually contain 3 degrees of freedom i.e., two signal powers in orthogonal polarizations and the inter-polarization phase difference. Therefore, one can in principle construct a SV-DD receiver to support 3-dimensional modulation formats and dramatically increase overall transmission capacities. Recently, a variety of SV-DD configurations supporting 2-, 3- and even 4-dimensional modulations are proposed and intensively studied.

2-dimensional (2D) PDM-DD systems: Morsy-Osman *et al.* proposed a simple 2D PDM-DD system in which two external intensity modulators (IM) are used to modulate two single-carrier PAM data streams onto two orthogonal polarizations as depicted in Fig. 20(a) [73]. At the receiver side, an SV-DD receiver (see Fig. 20(c)) is implemented by a polarization beam splitter (PBS) followed by 3 dB couplers to further split the signal giving four outputs. Ports 1 and 4 are fed into two PDs directly to obtain $|E_{r,x}|^2$ and $|E_{r,y}|^2$ which are used to calculate the SV components $S_{r,0} = |E_{r,x}|^2 + |E_{r,y}|^2$ and $S_{r,1} = |E_{r,x}|^2 - |E_{r,y}|^2$ in the digital domain, respectively. Ports 2 and 3 are fed into a balanced receiver consisting of a 90 degree hybrid with two balanced PDs (BPDs) to produce $S_{r,2}$ and $S_{r,3}$.

The subsequent DSP blocks for polarization de-multiplexing and equalization are shown in Fig. 21. In the 2D PDM-IM-DD systems, the PDM IM signals can be easily reconstructed

from the estimated transmitted Stokes parameters $\hat{S}_{t,0}$ and $\hat{S}_{t,1}$. Since $\hat{S}_{t,0}$ is insensitive to polarization rotation, it can be filtered with simple real-valued FIR filter $\mathbf{h}_{00}$ to mitigate residual ISI and obtain the estimate $\hat{S}_{t,0} = \mathbf{h}_{00} \otimes S_{r,0}$ where $S_{r,0}$ denotes the vector of the received Stokes parameter $S_{r,0}$ in time. But for the 1st SV component, it will suffer from the influence of polarization rotation. In order to recover $S_{t,1}$, a 3×1 multiple-input and single-output (MISO) bank of real-valued FIR filters $\mathbf{h}_{11}$, $\mathbf{h}_{21}$ and $\mathbf{h}_{31}$ is necessary. In this case, the estimated $\hat{S}_{t,1}$ is given by $\hat{S}_{t,1} = \mathbf{h}_{11} \otimes S_{r,1} + \mathbf{h}_{21} \otimes S_{r,2} + \mathbf{h}_{31} \otimes S_{r,3}$. The filters taps can be initialized using training symbols and LMS and switch to decision-directed mode upon convergence. In this case, the $n + 1$ th iteration of the filter estimate is given by

$$\begin{aligned} \mathbf{h}_{00}^{(n+1)} &= \mathbf{h}_{00}^{(n)} + \mu (e_x + e_y) \mathbf{S}_{r,0} \\ \mathbf{h}_{11}^{(n+1)} &= \mathbf{h}_{11}^{(n)} + \mu (e_x - e_y) \mathbf{S}_{r,1} \\ \mathbf{h}_{21}^{(n+1)} &= \mathbf{h}_{21}^{(n)} + \mu (e_x - e_y) \mathbf{S}_{r,2} \\ \mathbf{h}_{31}^{(n+1)} &= \mathbf{h}_{31}^{(n)} + \mu (e_x - e_y) \mathbf{S}_{r,3} \end{aligned} \quad (26)$$

where μ is the step size parameter, $e_{x(y)} = |\hat{E}_{t,x(y)}|^2 - |E_{t,x(y)}|^2$ are the error signals of the x (y) polarization.

An alternative 2D PDM-IM-DD scheme is proposed by Zhou *et al.* [74]. Here, the transmitter is the same as shown in Fig. 20(a) while at the receiver, the PDM-IM signal is divided into two different SOPs with a fixed $\pi/4$ polar angle difference and each of SOP signals is then split into two arbitrary but orthogonal polarization components by a PBS. The four PBS outputs are detected using four single-ended PDs as shown in Fig. 20(d) and are given by

$$\begin{bmatrix} |E_{r,x}|^2 \\ |E_{r,y}|^2 \\ |E_{r,x'}|^2 \\ |E_{r,y'}|^2 \end{bmatrix} = \begin{bmatrix} 1 + \cos \theta & 1 - \cos \theta & -\sin \theta \cos \varepsilon \\ 1 - \cos \theta & 1 + \cos \theta & \sin \theta \cos \varepsilon \\ 1 - \sin \theta & 1 + \sin \theta & -\cos \theta \cos \varepsilon \\ 1 + \sin \theta & 1 - \sin \theta & \cos \theta \cos \varepsilon \end{bmatrix} \cdot \begin{bmatrix} |E_{t,x}|^2 \\ |E_{t,y}|^2 \\ 2\text{Re}\{E_{t,x}E_{t,y}^*\} \end{bmatrix} \quad (27)$$

In contrast with the previous SV-DD receiver, only three output components $|E_{t,x}|^2$, $|E_{t,y}|^2$ and $2\text{Re}\{E_{t,x}E_{t,y}^*\}$ (i.e., $S_{r,2}$) can be extracted from the four received components. In this case, $S_{r,3}$ will be lost and $S_{r,2}$ can be seen as an unwanted mixed polarization beat interference (MPBI) term that needs to be removed. Since the MPBI product changes with the SOP, one can first estimate the rotation angle θ from (28) based on the ratio of power difference between the two pairs of PBS outputs i.e.,

$$\hat{\theta} = \tan^{-1} \left(\frac{\sum_k |E_{r,x}[kT_S]|^2 - \sum_k |E_{r,y}[kT_S]|^2}{\sum_k |E_{r,x'}[kT_S]|^2 - \sum_k |E_{r,y'}[kT_S]|^2} \right). \quad (28)$$

Based on the estimated angle $\hat{\theta}$, the MPBI can be removed by algebraic manipulations of the four equations in (29). As an example, one can calculate the intermediate variables $C_{x(y)} =$

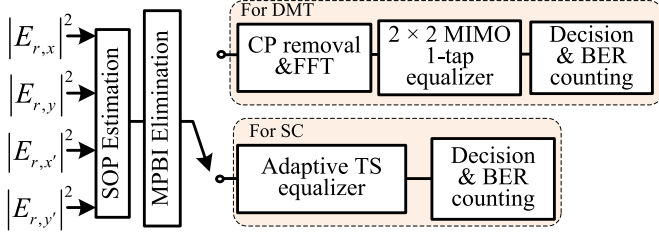


Fig. 22. DSP block diagrams for single-carrier (SC) PAM and multi-carrier DMT systems. TS: Training symbols; CP: Cyclic prefix; MPBI: Mixed polarization beat interference.

$|E_{r,x(y)}|^2 - |E_{r,x'(y')}|^2 \cdot \tan \hat{\theta}$ so that

$$\begin{bmatrix} C_x \\ C_y \end{bmatrix} = \begin{bmatrix} \cos \frac{\hat{\theta}}{2} & -\sin \frac{\hat{\theta}}{2} \\ \sin \frac{\hat{\theta}}{2} + \cos \frac{\hat{\theta}}{2} & \cos \frac{\hat{\theta}}{2} - \sin \frac{\hat{\theta}}{2} \\ -\sin \frac{\hat{\theta}}{2} & \cos \frac{\hat{\theta}}{2} \\ \cos \frac{\hat{\theta}}{2} - \sin \frac{\hat{\theta}}{2} & \sin \frac{\hat{\theta}}{2} + \cos \frac{\hat{\theta}}{2} \end{bmatrix} \begin{bmatrix} |E_{t,x}|^2 \\ |E_{t,y}|^2 \end{bmatrix} \quad (29)$$

and the transmitted signals $|E_{t,x}|^2, |E_{t,y}|^2$ can be obtained by matrix inversion. This is followed by ISI compensation. Note that single-carrier PAM or multi-carrier DMT signaling are equally applicable in PDM-DD systems as long as appropriate subsequent DSP blocks are implemented accordingly as shown in Fig. 22. For PAM signals, a bank of 4 FIR filters with real tap weights arranged in a butterfly structure can be used. The LMS algorithm can also be adopted to update the filter taps initially with training mode and switching to decision-directed mode upon convergence. For DMT signals, frequency-domain one-tap equalizers are used after cyclic prefix removal and FFT operation. Note that this scheme uses four PDs which can provide better RS than the 90 degree hybrids in SV-DD receiver shown in Fig. 20(c).

Moreover, both the aforementioned PDM-IM-DD systems are susceptible to the CD-induced frequency-selective fading which is important for inter-data center transmission. To this end, Che *et al.* proposed a PDM signal-carrier DD (PDM-SC-DD) system [75]. The transmitter configuration is shown in Fig. 20(b). The x -polarization is modulated with a complex signal using IQ modulator and a constant carrier A is sent on y -polarization. In this scheme, the transmitted SV is

$$\begin{bmatrix} S_{t,0} \\ S_{t,1} \\ S_{t,2} \\ S_{t,3} \end{bmatrix} = \begin{bmatrix} |E_{t,x}|^2 + |A|^2 \\ |E_{t,x}|^2 - |A|^2 \\ 2\text{Re}\{E_{t,x}A^*\} \\ 2\text{Im}\{E_{t,x}A^*\} \end{bmatrix} \quad (30)$$

The SV-DD receiver is the same as shown in Fig. 20(c). To first estimate the 3×3 polarization rotation matrix $\mathbf{R}$ for $[S_{r,1} S_{r,2} S_{r,3}]$ three sets of training symbols with the Jones space representation of $[0 \ 1]$, $[1 \ 1]$ and $[i \ 1]$ are sent at transmitter, which correspond to three time-interleaved training symbols $[-1 \ 0 \ 0]$, $[0 \ 1 \ 0]$ and $[0 \ 0 \ 1]$ in Stokes space, respectively. In this

TABLE V
KEY SIMULATION PARAMETERS FOR COMPARISON OF 2D PDM-DD SYSTEMS

Parameter	Values	Parameter	Values
Baud rate	56 Gbaud	ADC sample rate	80 G Sam/s
Laser linewidth	5 MHz	PD responsivity	0.65 A/W
Laser RIN	-160 dB/Hz	PD thermal noise	20pA/Hz ^{0.5}
Tx/Rx 3dB-bandwidth	20 GHz	PD dark current	10 nA

case, the estimation of first column of $\mathbf{R}$ is given by

$$\begin{bmatrix} S_{r,1} \\ S_{r,2} \\ S_{r,3} \end{bmatrix} = \underbrace{\begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix}}_{\mathbf{R}} \cdot \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{cases} \hat{h}_{11} = -S_{r,1} \\ \hat{h}_{21} = -S_{r,2} \\ \hat{h}_{31} = -S_{r,3} \end{cases} \quad (31)$$

and the second and third columns of $\mathbf{R}$ are estimated by the second and third pilot symbols in a similar fashion. Alternatively, we can use the blind least squares plane (LSP) estimation method that minimizes the sum of the distances between the plane and the received SVs [76]. In particular, after determining the normal vector (g_1, g_2, g_3) of the LSP, the polar angle θ can be estimated by

$$\hat{\theta} = \tan^{-1} \left(\sqrt{g_2^2 + g_3^2}, g_1 \right) \quad (32)$$

with the azimuth angle estimate

$$\hat{\varepsilon} = \tan^{-1} (g_3/g_1) \quad (33)$$

and the rotation matrix $\mathbf{R}$ can be obtained accordingly. This blind estimation algorithm can work for all modulation formats and offer performance close to training symbol based method. Then we multiply the inverse of $\mathbf{R}$ to estimate the transmitted SV and the original complex signal $\hat{E}_{t,x} = \hat{s}_{t,2} + j\hat{s}_{t,3}$. Subsequently, appropriate equalizers for single-carrier or multi-carrier signals similar to the one in Fig. 22 (but without the MIMO filter structure) are used to compensate both the additional ISI resulting from the low-pass filtering and other channel distortions.

We compared the transmission performance of the three 2D PDM-DD systems by simulations. For the same bit rate, PAM-4 is employed for PDM-IM-DD systems, and 16QAM is employed for PDM-SC-DD system. The simulation parameters are summarized in Table V and are based on commercially available components.

Fig. 23 gives the BER vs. received optical power performance of the three systems for back-to-back case. Results show that at the 7% FEC BER threshold, the received optical power for the three systems are -3 dBm for PDM-PAM4-DD(Hybrid); -4.2 dBm for PDM-PAM4-DD (4PDs) and -5.7 dBm for PDM-SC-16QAM-DD. Although PDM-SC-16QAM-DD system receiver structure is complex, it can achieve a better performance due to the higher sensitivity for 16QAM signal. Because the PDM-PAM4-DD (Hybrid) system is more complex than PDM-PAM4-DD (4PDs), it needs more PDs and an extra hybrid which bring more noise in the receiver causing lower RS.

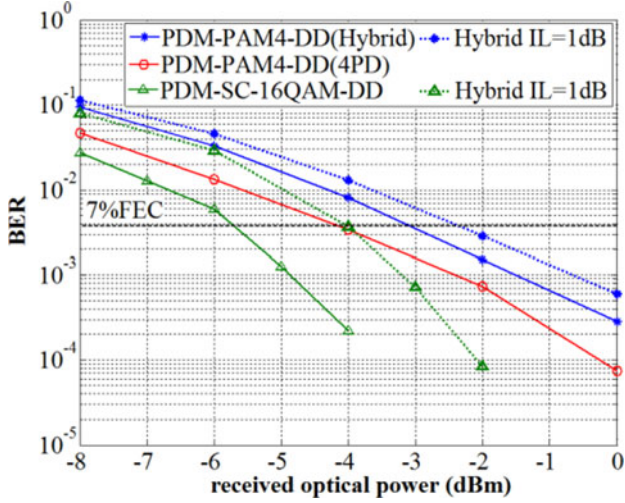


Fig. 23. Back-to-back 224 Gb/s transmission performance for various 2D PDM-DD systems.

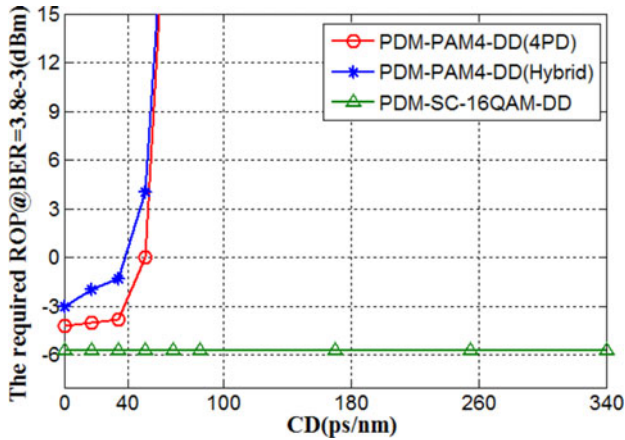


Fig. 24. The required ROP vs. CD for three 2D PDM-DD systems.

Considering the hybrid IL of around 1 dB, the RS of PDM-SC-16QAM-DD and PDM-PAM4-DD (Hybrid) system was deteriorated by 1.8 dB and 0.6 dB. This is because PDM-SC-16QAM-DD system uses hybrid outputs $S_{r,2}$ and $S_{r,3}$.

Next, the influence of CD is investigated for the three 2D PDM-DD systems. Fig. 24 shows the required ROP at a BER of 3.8×10^{-3} as a function of the accumulated CD from 0 ps/nm to 340 ps/nm which is equivalent to a SSMF in C-band of length up to 20 km. It can be seen that PDM-SC-16QAM-DD system is tolerant to CD effects while other systems quickly degrade with increasing CD. Hence, these two systems may not be suitable for ER C-band transmissions. Table VI summarizes and compares the three 2D PDM-DD systems.

3-dimensional (3D) PDM-DD systems: A 3D PDM-DD system is proposed in [77] with intensity modulation on two polarizations and an additional modulation of the phase difference ϕ between the two polarizations using a phase modulator (PM) as depicted in Fig. 25. A SV-DD receiver same as shown in Fig. 20 is used. In this case, information is encoded in all three degrees of freedom of Stokes space. The receiver DSP procedures for polarization de-multiplexing are shown in Fig. 26.

TABLE VI
COMPARISON OF 2D PDM-DD SCHEMES

2D Scheme	Hardware components	Cost	RS	CD tolerance	Application scenarios
PDM-PAM4-DD(Hybrid)	2 IMs+ 1 Hybrid.+ 6PDs	medium	low	low	DC links <20km
PDM-PAM4-DD (4PD)	2 IMs+4 PDs	low	medium	low	DC links <20km
PDM-SC-16 QAM-DD	1IQ+1Hybrid+6PDs	high	high	high	ER, metro access>40km

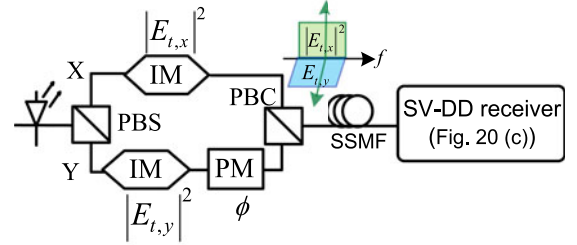


Fig. 25. 3D PDM-DD system transmitter using intensity modulator and phase modulator, and receiver architectures [77], [78].

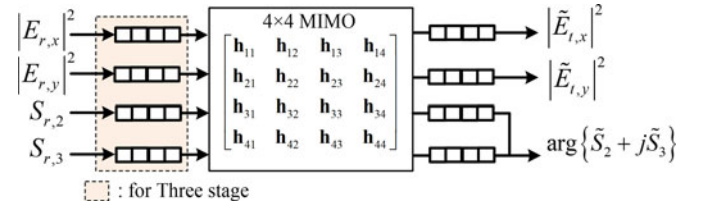


Fig. 26. The receiver DSP structure for polarization de-multiplexing for 3D PDM-DD systems.

In a two-stage implementation [78], the first stage of the DSP stack is a 4×4 MIMO that inverts the polarization rotation and mitigates mainly the receiver-side ISI. The equalizer taps are initialized using training symbols before switching to a decision-directed mode. The second stage comprises of four FIR filters to remove any residual ISI uncompensated by the first stage as well as transmitter-side residual ISI. As typical short-reach systems undergo considerable low-pass filtering effects, all 16 FIRs in the 4×4 MIMO filter bank have large number of taps which cause a significant increase of DSP complexity and longer convergence time. To address this issue, another implementation adds an additional stage of FIR [77] prior to the 4×4 MIMO. In this case, the first stage are four separate FIRs to mitigate ISI so that the subsequent 4×4 MIMO filter bank only requires a small number of taps for polarization de-multiplexing. In this case, the pre-filters can divert ISI compensation pressure away from 4×4 MIMO filter band so as to reach lower computational complexity. Then, the third stage mitigates residual transmitter side ISI for the two intensity and phase waveforms. Both the two-stage and three-stage approaches perform equally well but the three-stage DSP has much less computational complexity and can track the SOP faster with smaller latency as the tasks of ISI mitigation and polarization de-multiplexing are separated.

Li *et al.* [79] proposed an alternative 3D PDM-DD configuration in which two OFDM-16QAM signals are modulated on two

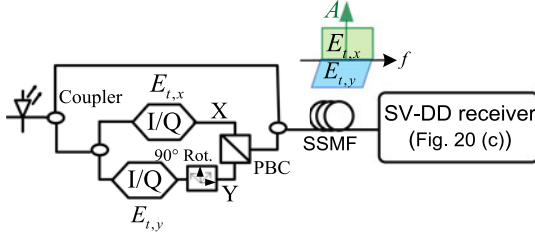


Fig. 27. 3D PDM-DD system transmitter using IQ modulators, and receiver architectures [79].

orthogonal polarizations using a dual-polarization IQ modulator with a SV-DD receiver. Then, the PDM signal are combined with a carrier A on x -polarization, as depicted in Fig. 27. Here, in order to obtain the linear mapping term of the fading-free complex signal and eliminate the SSBI impairment in x -polarization, the block-wise signal phase switching (SPS) scheme is employed for the x -polarization signal [80], which contains two consecutive blocks (in time domain) with the same data but their phases are with 90° difference. In this case, the spectral efficiency on x -polarization is reduced by half. The signals in both polarizations are then combined with the carrier A on x -polarization. At the receiver, training assisted or blind algorithms [76] can be used to estimate the transmitted SVs, i.e., $\hat{S}_{t,0}$, $\hat{S}_{t,1}$, $\hat{S}_{t,2}$ and $\hat{S}_{t,3}$. The transmitted signal in x -polarization can then be recovered by

$$\left| \hat{E}_{t,x} + A \right|^2 = \left(\hat{S}_{t,0} + \hat{S}_{t,1} \right) / 2 \quad (34)$$

Once the signal on x -polarization, namely $A\hat{E}_{t,x}$, is fully recovered, the signal on y -polarization signal can be extracted from the SV components of $\hat{S}_{t,2}$ and $\hat{S}_{t,3}$. Because

$$\hat{S}_{t,2} + j\hat{S}_{t,3} = 2A\hat{E}_{t,y}^* + 2\hat{E}_{t,x}\hat{E}_{t,y}^* \quad (35)$$

the y -polarization signal is obtained by

$$\hat{E}_{t,y} = \left(\left(\hat{S}_{t,2} + j\hat{S}_{t,3} \right) / \left[2 \left(\hat{E}_{t,x} + A \right) \right] \right)^* \quad (36)$$

Note that this system can mitigate CD-induced power fading effects while enabling 3D modulation at the expense of a complex DSP and potentially large latency.

In particular, estimating the polarization rotation matrix and recovering the transmitted SVs are similar to the 3D PDM-DD systems discussed earlier. However, demodulation of inter-polarization differential phase requires the estimated polarization rotation matrix to be fed back to the transmitter to pre-rotate the transmitted waveforms so that $E_{t,x(y)}(t)$ is polarization aligned with $E_{r,x(y)}(t)$ and the two BPD outputs of the second 90 degree hybrid in Fig. 28 will then give

$$\begin{aligned} S'_{t,2} &= 2\text{Re} \left\{ \hat{E}_{t,x} [kT_S] \hat{E}_{t,y}^* [(k-1)T_S] \right\} \\ S'_{t,3} &= 2\text{Im} \left\{ \hat{E}_{t,y}^* [kT_S] \hat{E}_{t,x} [(k-1)T_S] \right\}. \end{aligned} \quad (37)$$

The drawback of this system is the need for the rotation matrix to be fed back to the transmitter to pre-rotate the transmitted waveforms. Although in principle the receiver DSP can track

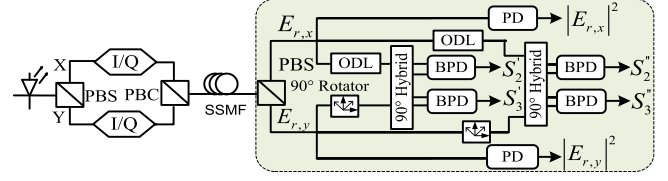


Fig. 28. 4D PDM-DD system transmitter and receiver architectures. ODL: Optical delay line (by a symbol period T_s).

the SOP using the 4×4 MIMO DSP running on the first three DOFs and continuously update the rotation matrix that is fed back to the transmitter, this approach may suffer from relatively slow tracking speed due to the feedback delay especially in fast varying SOP scenarios. Recently, T. M. Hoang *et al.* [81] proposed a simpler 4D- PDM-DD scheme by using a SV-DD receiver combined with KK technique. A 400 Gbit/s net bit rate transmission over 80 km SSF at 1550 nm was demonstrated. Table VII summarizes and compares various PDM-DD systems proposed to date.

The PDM-DD technologies are seen as possible approaches to replace PDM coherent detection to increase spectral efficiencies in short-reach communications. Comparing with coherent detection, PDM-DD does not require a LO laser and has a lower linewidth requirement on transmitter laser. However, exploiting all dimensions for information transmission comes with additional hardware complexity. For instance, a SV-DD receiver can support 3-D transmissions but it is much more complex compared to single polarization IM-DD systems with higher cost and higher IL, which directly translate into penalty on RS or OSNR. Therefore, further investigations are much needed to explore the tradeoffs between spectral efficiency, ISI, algorithmic and hardware complexity and determine the best solution for specific short-reach application scenarios.

VI. DEVICES ADVANCES FOR SHORT-REACH SYSTEMS AND ASSOCIATED DSP

As discussed in the Introduction, short-reach systems hinge on new and low-cost devices and appropriate DSPs are intricately related to the devices characteristics. We discuss a few key trends in devices developments for short-reach systems.

Single mode (SM) VCSELs: One appealing solution for short-reach systems is based on short wavelength (850 nm) VCSELs and MMFs that has been widely deployed in the past for 10 Gb/s class optical inter connections. However, high speed optical inter-connections with bit rates up to 100 Gb/s or even 400 Gb/s and reach up to 300 m or longer are desired in the future. The main stumbling block is modal and chromatic dispersion in MMFs. Although the introduction of high bandwidth FMFs (OM4) helps, it is still difficult to achieve the desired performance with conventional VCSELs. One effective way is to reduce the number of transverse modes of VCSELs to achieve SM operation. R. Safaisini *et al.* integrated a mode filter with a VCSEL which suppressed higher order transverse modes and successfully demonstrated 25 Gbit/s NRZ transmission over 1.3 km of OM4 fiber [96]. Alternatively, SM VCSELs can be realized with the leakage of higher order transverse modes via

TABLE VII
KEY PDM-DD SYSTEMS DEMONSTRATED WITH 2-, 3- AND 4-DIMENSIONAL MODULATIONS

Dim.	Modulation formats	Bit rate(Gbit/s)	bits/symbol	Wavelength (nm)	Reach (km)	FEC	Receiver devices	Ref.
2D	2 PAM4	224/168	4	1310	10	3.8e-3	2PD+2BPD+H	[73] [95]
	32-QAM	125	5	1550	160	3.8e-3	2PD+2BPD+H	[83]
	OFDM-16QAM	160	4	1550	160	2e-2	2PD+2BPD+H	[75]
	2 DMT-16QAM	200	4	1550	1	3.8e-3	2PD+2BPD+H	[84]
	16-QAM	40	4	1550	160	3.8e-3	2PD+2BPD+H	[76][85]
	10×16-QAM	1024	4	1550	480	2e-2	2PD+2BPD+H	[86][89]
	OFDM-64QAM	80	6	1550	160	2e-2	4PD(2PD+2BPD)+H	[87][88]
3D	16QAM/8QAM/QP SK	224/168/112	4/3/2	1550	320/1000/2880	3.8e-3	2PD+2BPD+H	[92]
	2 PAM4+4PM	336	6	1550	10	3.8e-3	2PD+2BPD+H	[77][94]
	16-QAM+16-QAM +C	192	6	1550	160	3.8e-3	3BPD+H	[79]
	16-QAM-PAM2	280	6	1550	320	2e-2	2PD+2BPD+H	[90]
4D	2 PAM4+4PM	504	6	1550	0.5	3.8e-3	2PD+2BPD+H	[91]
	2-ring/8PSK	320/360	8	1310	10/20	3.8e-3/	2PD+4BPD+2-H	[82][93]
	16QAM	400	8	1550	80	2e-2	2PD+2BPD+H	[81]

PD- single-ended photo-detector; BPD – balanced photo-detector; H: 90 degree hybrid

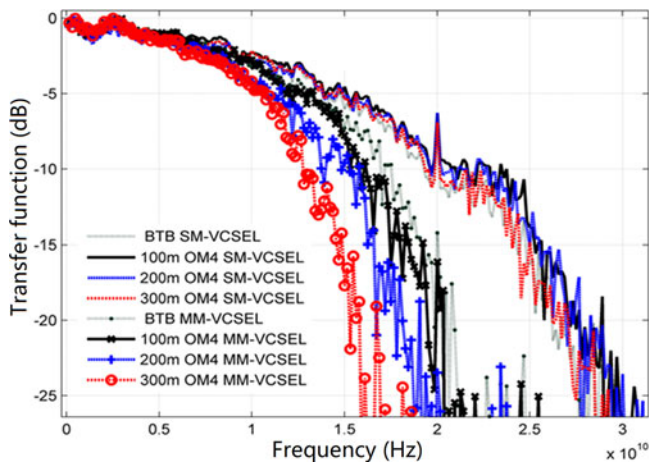


Fig. 29. End-to-end frequency response of an optical transmission system using MM-VCSEL and SM-VCSEL [98].

oxide aperture engineering [97]. Since only one transverse mode exist in SM-VCSEL, inter-modal dispersion in MMFs would be absent for SM-VCSELs. Using this SM-VCSEL, Bo *et al.* experimentally demonstrated 100 Gbit/s DMT transmission over 300 m OM4 fiber [98] as shown in Fig. 29. Puerta *et al.* experimentally demonstrated 107.5 Gbit/s transmission over 100 m of OM4 fiber using SM-VCSEL while only 10 m transmission over OM4 can be achieved using MM-VCSEL [99]. In 2017, C. Kottke *et al.* experimentally demonstrated a record transmission of 113 Gbit/s over 550 m OM4 fiber below HD-FEC using SM-VCSEL with DMT modulation [100].

Double-side EML: EMLs are widely used in short-reach links due to its well-developed technology, small form factor, high bandwidth and large extinction ratio. Normally, EMLs have only one modulated output, while the other output is used for monitoring. In 2016, Theurer *et al.* proposed double side EML (DS-EML) which integrates two EAMs on both sides of the

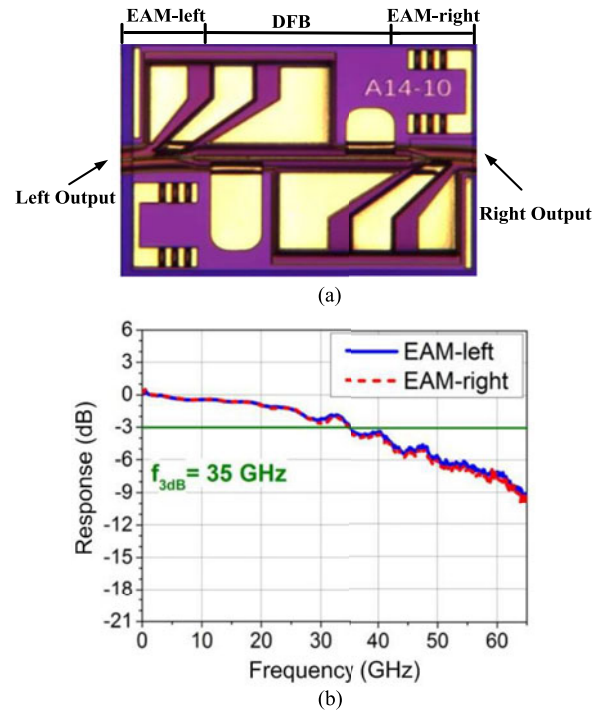


Fig. 30. (a) Microscopic view of the DS-EML chip. (b) Frequency response of both DS-EML sides. Current = 70 mA, bias voltage = -1 V, $T = 20$ °C.

laser. DS-EML provides two independent optical modulated signals at the same wavelength. Fig. 30(a) shows the structure of a DS-EML. It comprises a distributed feedback (DFB) laser with EAMs located on both sides of the laser. The emission wavelength of the DFB laser is in the C-band. More details of the DS-EML can be found in [101]. Fig. 30(b) shows the frequency response of both DS-EML sides with an injected current of 70 mA and a bias voltage of -1 V at 20 °C. A 3 dB bandwidth of 35 GHz was demonstrated for both sides of DS-EML.

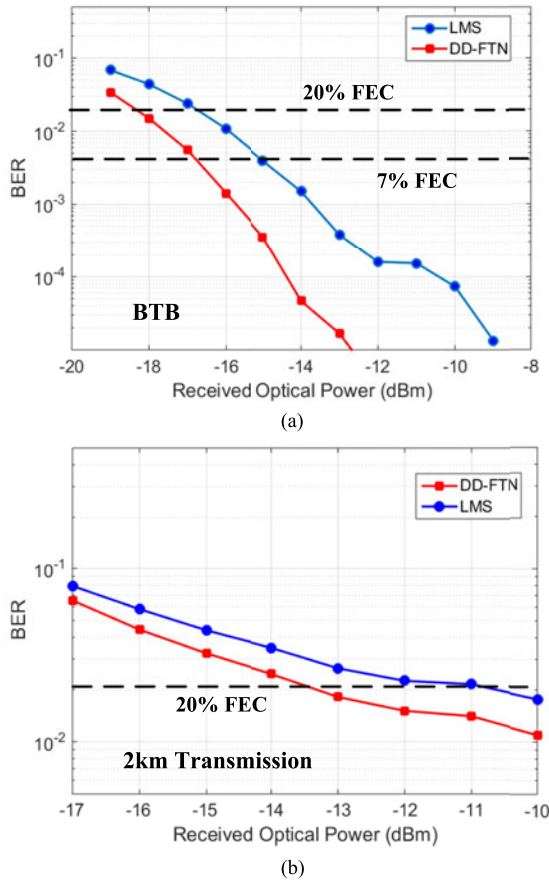


Fig. 31. BER vs. received optical power for 120 GBaud PDM-NRZ signal: (a) back-to-back transmission, (b) after 2 km transmission [103].

The output power from EML chip is around 3 dBm for each side. Due to imperfect coupling during the packaging process, the output powers of the DS-EML module were measured to be -1 dBm and -0.8 dBm for the right and left outputs of the DS-EML, respectively, with a bias voltage of 0 V and an injection current of 70 mA. M. Theurer *et al.* experimentally demonstrated generation of 2×56 Gbit/s NRZ signal from single a DS-EML chip for multi-lane short reach applications, which reduce the total cost by half for multi-lane applications. Meanwhile, a 56 Gbit/s optical PAM4 signal was experimentally generated by combining two optical NRZ signals from DS-EML and transmitted over 10 km of fiber with a receiver sensitivity of -14.2 dBm. This PAM4 generation method is based on the unique features of DS-EML that only requires binary electrical driving signal and avoid the need for high speed DAC and linear electrical amplifier for PAM-4 signal generation [102]. Furthermore, as it is difficult to generate PDM signals with two discrete EMLs due to relative wavelength drifting, DS-EML is ideal for PDM signal generation as it has two independent modulated outputs at the same wavelength. To this end, Zhong *et al.* successfully generated a 120 GBaud PDM-NRZ signal using DS-EML [103].

Fig. 31(a) shows the measured BER as a function of received optical power for 120 GBaud PDM-NRZ signal for back to

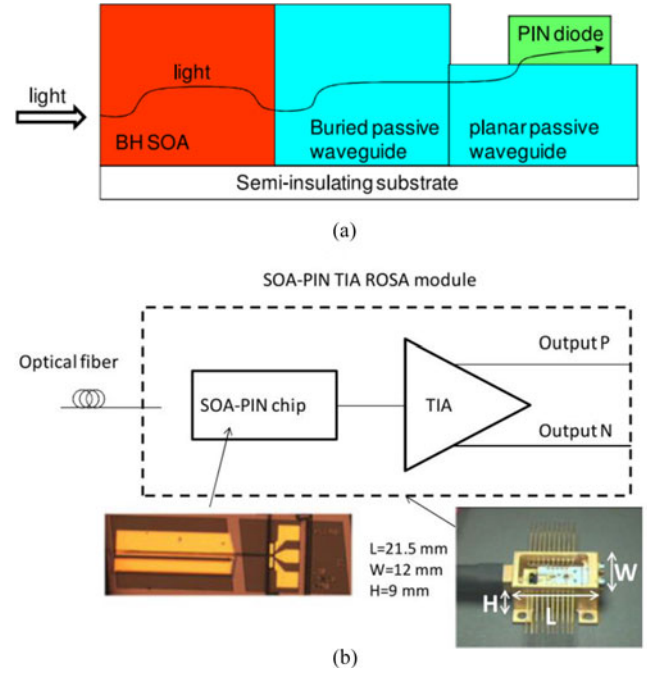


Fig. 32. Schematic view of (a) SOA-PIN chip and (b) packaged SOA-PIN/TIA receiver [106]. BH SOA: Buried heterostructure semiconductor optical amplifier.

back transmission. It can be seen that DD-FTN can significantly improve the BER performance. RS at the 7% over head FEC limit of $3.8E-3$ are -15 dBm and -17 dBm for linear equalization and DD-FTN, respectively. Fig. 31(b) shows the BER as a function of received optical power for 120GBaud PDM-NRZ signal transmitted over 2 km of SSMF. In this case, performance can only reach 20% overhead FEC limit of $2E-2$ due to impact of CD. Nonetheless, an improvement of 2.6 dB in terms of RS was demonstrated using DD-FTN. The impact of CD can be mitigated by operating at O band or employing other CD compensation techniques. The DS-EML is potentially a good device for future high speed short reach applications. Further studies and demonstrations based on DS-EML are currently under way.

Semiconductor optical amplifier (SOA)-PIN/TIA receiver: To achieve the same the target BER, a higher SNR is required for higher bit rate systems, which results in a reduction in RS. In order to achieve high RS or high-power budget, optical pre-amplification is necessary. EDFA can be adopted but it leads to larger footprints and high costs, which are not suitable for short-reach systems. To reduce the size and cost, Mason *et al.* demonstrated a monolithic integrated SOA-PIN/TIA receiver which has a RS of -17.2 dBm at BER of $1E-1$ for 40 Gb/s [104]. However, this receiver only operates in TE polarization. C. Caillaud *et al.* recently demonstrated an integrated SOA-PIN chip based receiver with low polarization dependent gain (PDG) [105]. Fig. 32(a) shows the schematic of the SOA-PIN chip. The chip used a buried heterostructure SOA (BH-SOA) and a deep ridge evanescent PIN photodiode made on a semi-insulating InP substrate. SOA and PIN are linked by a passive waveguide which comprises a buried section and a planar section [106]. Fig. 32(b) shows a schematic view of the SOA-PIN/TIA receiver module, which has a responsivity of 44 A/W, a polarization dependent

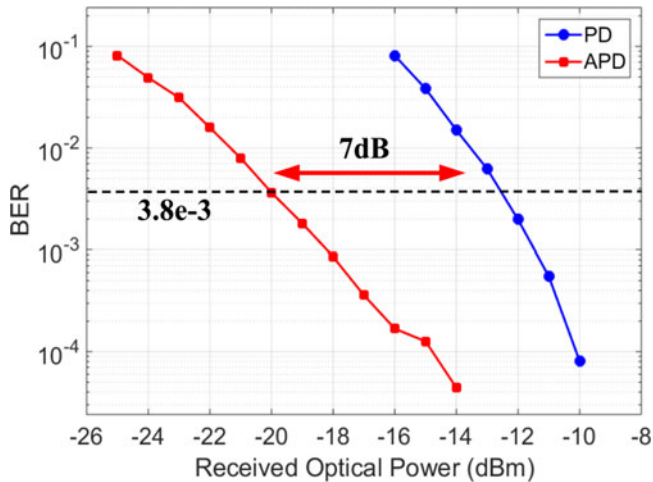


Fig. 33. Measured BER as a function of received optical power for back-to-back transmission with APD (red curve) and PD (blue curve) [111].

loss below 2 dB, noise figure of 8.5 dB and a 3 dB bandwidth of 35 GHz. Record sensitivities (at $\text{BER} = 10^{-9}$) of -23 dBm and -21 dBm were achieved for 25 Gb/s and 40Gb/s NRZ transmission systems.

High bandwidth APD receiver: As APD has much higher responsivity than PIN diode, it is a promising device to achieve low cost, high RS and long reach transmission. However, for a long period of time, there is a lack of high bandwidth APD for high speed applications. In 2012, M. Nada *et al.* reported an APD-ROSA with a 3-dB bandwidth of 18GHz for 25 Gbit/s operations at 1310 nm. A RS (at $\text{BER} = 10\text{E-}12$) of -21.7 dBm is demonstrated at a data rate of 25.8 Gbit/s [107]. Using this APD-ROSA, M. Nada *et al.* experimentally demonstrated optical amplifier-less transmission of 56 Gbit/s NRZ and PAM-4 over 40 km at 1310 nm [108], [109]. In 2016, M. Huang demonstrated a high bandwidth Ge/Si APD with a record sensitivity (at $\text{BER} = 2\text{E-}4$) of -23.5 dBm for 25 Gbps NRZ signal at 1310 nm [110]. Since it can be easily integrated with other silicon photonics devices, this Ge/Si APD is a promising device for future high-speed data center applications. In 2017, K. Zhong *et al.* experimentally demonstrated an amplifier-less transmission of 56 Gbit/s PAM4 signal over 60 km using 25G EML and Ge/Si APD at O band [111]. Fig. 33 shows the measured BER as a function of received optical power for back-to-back transmission with APD and PD, respectively. RS at 7% overhead FEC limit of $3.8\text{E-}3$ is -20 dBm for an APD system. A RS of -13 dBm was demonstrated for PD system at 7% overhead FEC limit of $3.8 \text{E-}3$. An improvement of 7 dB in terms of RS was achieved using APD [111]. Recently, K. Zhong *et al.* experimentally demonstrated for the first time an amplifier-less transmission of a single lane 112 Gbit/s PAM-4 signal over 40 km using 25 G EML and an APD at O band with a record RS (at $\text{BER} = 3.8\text{E-}3$) of -14.8 dBm [112]. There will be more demand for high bandwidth and high responsivity APDs for short-reach optical inter-connects in the future.

Integrated transmitter and receiver structures: For many of the schemes discussed above, the transmitter and receiver consist of a number of discrete components. To reduce system

form factors and ultimately cost, system integration based on silicon photonics technology is a promising solution. This will enable simple PAM signal generation, CD pre-compensation and WDM multiplexing to be implemented within a simple integrated transmitter. Much research efforts have been made in this direction [113], [114]. Impressive results are reported for integrated silicon photonic PDM-DD transmitters and receivers [115].

Advanced fiber for short-reach applications: Besides VCSELs, another promising way to improve intra-data-center links is to increase the effective modal bandwidth (EMB) as a result of complex modal and chromatic dispersion interactions (MCDI) of MMF. The latest two MMFs right are OM3 and OM4 fibers, for which the EMBs are $2000 \text{ MHz} \cdot \text{km}$ and $4700 \text{ MHz} \cdot \text{km}$, respectively [116]. Today, new OM4 fibers can support an EMB up to $10 \text{ GHz} \cdot \text{km}$. Further capacity increases can be achieved using shortwave WDM based on a wideband MMF [117]. Y. Sun *et al.* experimentally demonstrated 51.56 Gb/s PAM4 transmission over 150 m WB-MMF using SWDM TOSA thus indicating that an aggregated 200 Gb/s data rate is achievable on a single WB-MMF [118]. Spatial mode multiplexing in FMFs has been studied and employed to increase the transmission capacity of long-haul systems in the past few years. In short-reach transmission systems, spatial mode multiplexing in FMFs can also be used to increase system capacity. However, the required MIMO process to compensate the modal cross-talk is especially undesirable for short-reach systems. In this connection, polarization maintaining elliptical ring core fiber [119] and elliptical core mode selective photonic lanterns [120] are recently reported to support MIMO-free MDM high capacity short-reach systems.

It should be noted that in contrast to developing better DSP, there are also attempts to design low-power DSP-free coherent receivers for data-center applications [121]. For PDM-QPSK signals, PDM can be achieved by marker tone detection circuitry and carrier recovery is achieved by multiplier-free phase lock loops. Finally, advances in silicon photonics are enabling light sources, modulators and photo-detectors to be integrated in the silicon integrated circuit infrastructure and is opening up tremendous opportunities for high-speed, low-cost and small form factor transceivers ideal for data center applications. Interested readers are referred to [122] for a review of recent developments of this exciting area of research that may play a key role in future short-reach systems.

VII. CONCLUSIONS AND OUTLOOK

In this paper, we have reviewed various DSP techniques that enable high capacity transmissions for short-reach systems. Advanced modulation formats and direct detection are typically used due to cost constraints. DSP techniques are essential to overcome component bandwidth limitations, impairments introduced by nonlinear modulation characteristics and CD in the transmission channel. Further system performance improvements can be obtained through SSB signaling with advanced interference cancellation techniques. Various approaches for realizing direct detected polarization division multiplexed transmissions are promising techniques for further increase in

system capacity. Practical realizations of many of the schemes discussed rely on the advancements in component technologies that improve receiver sensitivities and enable the integration of many transmitter and receiver functions. Future efforts on joint optimization of optical components and DSP techniques may become a key area of growth. This may also enable gradual introduction of many coherent detection functionalities into future short-reach optical communication systems.

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